SLIPPER ZERO+ : ESP32-BASED Wi-Fi PENETRATION AND DEFENSE TOOLKIT

PROJECT PHASE II REPORT

Submitted by

AROMAL H	**RIE22CC021**
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to

APJ Abdul Kalam Technological University

in partial fulfilment of the requirements for the award of the Degree of

Bachelor of Technology (B.Tech)

in

Computer Science and Engineering (Cyber Security)

Under the guidance of

Ms. BIJI BABU



Department of Computer Science and Engineering (Cyber Security)

Rajadhani Institute of Engineering and Technology (Autonomous)

Nagaroor, Thiruvananthapuram

April, 2026

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DECLARATION

We undersigned hereby declare that the final project report **"SLIPPER ZERO+ : ESP32-BASED Wi-Fi PENETRATION AND DEFENSE TOOLKIT**", submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of **Ms. BIJI BABU**, **Assistant professor, Department of Computer Science & Engineering**. This submission represents our ideas in our own words and where ideas or words of others have been included. We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Place: Nagaroor **Signature**

Date: April, 2026 Midhil V P

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
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CERTIFICATE

This is to certify that the report entitled "**SLIPPER ZERO+:ESP32-BASED Wi-Fi PENETRATION AND DEFENSE TOOLKIT"** submitted by **MIDHIL V P (RIE22CC041)** to the APJ Abdul Kalam Technological University for the partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ACKNOWLEDGEMENT

First and foremost, we express our heartfelt thanks and profound gratitude to the Management of Rajadhani Institute of Engineering and Technology and our Principal, **Dr. MADHUKUMAR S** for providing the opportunity to do this project during the final semester of our B.Tech degree course.

We are extremely happy to mention a great word of gratitude to **Dr. SANGEETHA SHIBU**, Head of department of Computer Science and Engineering , for providing us with all facilities for the completion of this work.

We would like to express our sincere gratitude to our guide **Ms. BIJI BABU**, for her relevant suggestions, guidance and inestimable leadership given for the completion of project phase 2 report. We thank our Project Coordinator **Ms. ASWATHY L C** for her constant encouragement during the entire project work.

Finally, we take this opportunity to express our gratitude to our parents for their love, care, and support, and also to our friends and well-wishers who have been constant sources of support and inspiration in completing the project work.

VISION OF THE INSTITUTE

Groom the Youth as Innovative, Creative and Empathetic Technologists, Hospitality Professionals, Managers and Entrepreneurs for Social Transformation.

MISSION OF THE INSTITUTE

M1: To encourage holistic development of students with well-balanced curricular, co-curricular and extra- curricular activities.

M2: To promote ethical and value-oriented teaching, research and consultancy among faculty and students for social transformation.

M3: To interact with industrial organizations, governmental agencies, engineering, hospitality and business enterprises for collaborative learning.

VISION OF THE DEPARTMENT

To be a premier department nurturing students to be strong in core and interdisciplinary knowledge, employable skills, industrially competent, innovative and creative professionals with ethical values.

MISSION OF THE DEPARTMENT

M1: To develop competency in Computer Science & Engineering through outcome-based education.

M2: To imbibe the culture of creativity and critical thinking to enhance research and innovations in advanced computing technology.

M3: To enhance leadership, problem solving and entrepreneurial skills to transform social responsibilities with ethical values.

PROGRAMME EDUCATIONAL OBJECTIVES

Graduates will

PEO 1: Accomplish knowledge, skills and proficiency in various domains of Computer Science & Engineering

PEO 2: Develop innovative computing products for reconcile contemporary issues in industry.

PEO 3: Demonstrate team work, ethical leadership and promote entrepreneurship.

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Graduates are able to

PSO 1: Able to analyze and design solutions for complex problems in computing technologies using software project management concepts and programming languages.

PSO 2: Develop proficiency in inter-disciplinary areas of computer engineering to perform tasks related to industry & research.

PSO 3: Competent to solve real world problems pertaining to product and services in Information Technology.

PROGRAMME OUTCOMES

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

COURSE OUTCOMES

Course Outcomes [COs]: After successful completion of the course, the students will be able to:

CO1	Identify academic documents from the literature which are related to her/his areas of interest (Cognitive knowledge level: Apply)
CO2	Read and apprehend an academic document from the literature which is related to her/ his areas of interest (Cognitive knowledge level: Analyze).
CO3	Prepare a presentation about an academic document (Cognitive knowledge level: Create).
CO4	Give a presentation about an academic document (Cognitive knowledge level: Apply).
CO5	Prepare a technical report (Cognitive knowledge level: Create).

CO MAPPING TO POs

Mapping of course outcomes with program outcomes

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	2	2	2	1	2	2	2	1	1	1	1	2
CO2	2	2	2		1	3	3	1	1		1	1
CO3									3	2	2	1
CO4					2			3	2	2	3	2
CO5	2	3	3	1	2							1

ABSTRACT

Wi-Fi communication has become an integral part of modern computing, enabling seamless connectivity across personal devices, enterprise systems, and IoT-based environments. Despite advancements in security protocols such as WPA2 and WPA3, wireless networks remain vulnerable to various attacks including deauthentication flooding, rogue access point creation, and unauthorized network access. These vulnerabilities arise due to weaknesses in wireless communication protocols and the lack of continuous monitoring mechanisms, making wireless environments a common target for cyber threats.Traditional penetration testing tools such as Aircrack-ng, Wireshark, and Reaver provide powerful capabilities for analyzing wireless networks and executing attacks. However, these tools require specialized hardware such as external Wi-Fi adapters, Linux-based configurations, and significant technical expertise. Additionally, most of these tools focus primarily on offensive testing and do not provide integrated real-time detection or monitoring capabilities. This makes them less suitable for beginners, educational environments, and scenarios where portability and simplicity are required.To address these limitations, this project presents Slipper Zero+, a low-cost and portable Wi-Fi penetration and defence toolkit developed using the ESP32 platform. The system integrates both offensive and defensive functionalities within a unified framework. The offensive capabilities include deauthentication attack simulation and controlled wireless packet injection, while the defensive mechanisms focus on detecting abnormal network behavior such as excessive deauthentication frames using a lightweight intrusion detection approach.The system also incorporates a web-based dashboard that provides real-time visualization of detected networks, connected devices, signal strength, and system activity. It displays alerts generated by the intrusion detection module and allows users to control system operations such as scanning and attack execution through an interactive interface. This eliminates the need for complex command-line tools and enables users to perform wireless security analysis in a simplified and accessible manner.By combining embedded hardware capabilities with real-time packet monitoring and visualization, Slipper Zero+ reduces the complexity associated with traditional penetration testing environments.

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LIST OF ABBREVIATIONS

AP	Access Point
IDS	Intrusion Detection System
HTTP	Hyper Text Tranfer Protocol
GUI	Graphical User Interface
SSID	Service Set Identifier
WPA	Wi-Fi Protected Access
WPA2	Wi-Fi Protected Access 2
WS	WebSocket
API	Application Programming Interface
GUI	Graphical User Interface
MAC	Media Access Control
TCP	Transmission Control Protocol
IoT	Internet of Things
JSON	JavaScript Object Notation
LED	Light Emitting Diode
Wi-Fi	Wireless Fidelity

CHAPTER 1

INTRODUCTION

1.1 Overview

Wi-Fi technology has become one of the most widely adopted modes of digital communication, enabling seamless wireless connectivity across personal devices, enterprise networks, and rapidly growing IoT infrastructures. With the exponential increase in smart devices and automated systems, Wi-Fi has evolved from a convenience-based communication medium into a critical component supporting real-time applications, remote accessibility, and interconnected environments. However, this widespread reliance has also introduced significant security challenges, as wireless networks are inherently more vulnerable to cyber threats compared to wired systems. Attacks such as deauthentication flooding, rogue access point creation, and unauthorized access attempts continue to exploit weaknesses in wireless communication protocols. Despite advancements in encryption standards such as WPA2 and WPA3, vulnerabilities within the IEEE 802.11 protocol and improper network configurations still expose wireless systems to potential exploitation.Traditional approaches to Wi-Fi security assessment rely heavily on software-based penetration testing tools such as Aircrack-ng, Wireshark, Kismet, and similar frameworks.

These tools require external wireless adapters capable of monitor mode and packet injection, along with Linux-based environments and specialized drivers. While they provide powerful capabilities for network analysis and attack execution, they demand significant computational resources and technical expertise. The configuration process often involves complex setup procedures, including driver installation, firmware modification, and compatibility verification. As a result, these tools are not easily accessible to beginners, students, or institutions with limited resources. Furthermore, most existing solutions focus primarily on offensive testing techniques without integrating real-time detection or monitoring capabilities, limiting their effectiveness in providing a complete understanding of wireless security.The proposed project, Slipper Zero+: ESP32-Based Wi-Fi Penetration and Defence Toolkit, is developed to address these limitations by offering a cost-effective, portable, and integrated solution for wireless security analysis. The system is built on the ESP32 microcontroller platform, which supports promiscuous mode for capturing wireless packets and raw packet injection for simulating attacks. The project combines both offensive and defensive functionalities within a unified framework. Offensive operations include deauthentication attack simulation and controlled wirelessinterference, while defensive mechanisms focus on detecting abnormal traffic patterns such as excessive deauthentication frames.

This dual capability enables users to both generate and analyze wireless threats within the same system.A key component of the system is the integration of a web-based dashboard, which provides real-time visualization and control of system operations. The dashboard displays detected access points, connected devices, signal strength, and network activity, allowing users to monitor the wireless environment effectively. It also presents alerts generated by the intrusion detection mechanism, enabling users to identify suspicious behavior immediately. By providing an interactive and user-friendly interface, the system eliminates the need for command-line interaction, making it accessible to users with varying levels of technical expertise. This feature significantly enhances usability and supports both learning and practical experimentation.

The system demonstrates that essential wireless security testing functionalities can be implemented on lightweight embedded hardware without requiring high-end systems or external adapters. By combining packet monitoring, intrusion detection, and attack simulation, the project provides a balanced approach to understanding wireless vulnerabilities. The ability to capture real-time data and visualize network behavior allows users to observe how wireless systems respond to different types of attacks. This practical approach improves the overall learning experience and provides insights into both offensive and defensive aspects of cybersecurity.Furthermore, the project contributes to improving accessibility in cybersecurity education and research. In many learning environments, hands-on experience with wireless penetration testing is limited due to the high cost of equipment and the complexity of setup procedures.

This system enables users to perform practical experiments using a compact and affordable device, without requiring extensive configuration or additional hardware. It provides a platform for students, researchers, and professionals to explore wireless security concepts, understand attack mechanisms, and evaluate detection strategies in a controlled environment.The project also addresses real-world challenges associated with IoT security, where a large number of devices rely on wireless communication for operation. Since IoT systems are often deployed in environments with limited security controls, they are particularly vulnerable to wireless attacks. By providing a tool capable of monitoring network activity, detecting anomalies, and simulating attacks, the system helps in identifying vulnerabilities at an early stage. This supports proactive security measures and reduces the risk of large-scale system compromise.

CHAPTER 2

LITERATURE SURVEY

2.1 ESP-Based Wi-Fi Security Analysis Systems

Author(s): Himanshu Tiwari et al., 2024

Overview:

This study explored the application of ESP8266 and ESP32 microcontrollers for performing Wi-Fi security analysis and penetration testing tasks. The research demonstrated that low-cost embedded platforms can be effectively used to execute various wireless attack techniques, including deauthentication attacks, beacon flooding, and rogue access point creation. The system leverages the promiscuous mode capability of ESP modules to capture wireless packets transmitted within the surrounding network environment.The study emphasized that ESP-based systems are capable of performing real-time packet monitoring and network analysis without the need for high-end computing devices. By utilizing lightweight firmware and embedded programming, the system was able to continuously scan nearby networks, identify access points, and analyze wireless traffic patterns. This approach significantly reduces the dependency on traditional penetration testing setups that require laptops, specialized Wi-Fi adapters, and complex configurations.

Furthermore, the research highlighted the growing importance of making cybersecurity tools more accessible and affordable, particularly for students, researchers, and educational institutions. The use of compact embedded devices enables the deployment of wireless security testing tools in real-world environments with minimal setup. The portability of the ESP platform allows users to carry out network analysis in different locations, making it suitable for field-based experimentation.

However, the study primarily focused on attack execution and did not incorporate defensive mechanisms such as intrusion detection or monitoring systems. It also lacked a structured user interface for visualizing network data, which limits usability for non-technical users. Despite these limitations, the research provided a strong foundation for developing integrated systems that combine both offensive and defensive capabilities in a single platform.

Limitations:

While the study demonstrates the effectiveness of ESP-based platforms for performing wireless security analysis, it also presents several limitations. The system primarily focuses on executing attack techniques and lacks the integration of defensive mechanisms such as intrusion detection or real-time monitoring systems. This restricts its capability to function as complete security solution.

Additionally, the absence of a structured and user-friendly interface limits accessibility for non-technical users, making it difficult to visualize network data and system operations effectively.The implementation also does not provide a unified framework that combines both offensive and defensive functionalities, resulting in limited system integration. Although the hardware platform is cost-effective and portable, the computational constraints of ESP modules may affect performance when handling complex analysis or large-scale network environments. Furthermore, the reliance on basic firmware-based operations restricts the system's ability to support advanced security features or scalable architectures required for enterprise-level deployment.

2.2 Wireless Network Attack Simulation Using Penetration Testing Tools

Author(s): J. Singh, 2023

Overview:

This research focused on the use of established penetration testing tools such as Aircrack-ng, Wireshark, and Reaver for analyzing wireless network vulnerabilities. The study demonstrated how these tools can be used to capture network packets, monitor wireless traffic, and perform attacks such as password cracking, packet injection, and network spoofing.The research provided a comprehensive understanding of how attackers exploit weaknesses in wireless security protocols. Tools like Wireshark enable detailed packet inspection, allowing analysts to examine network communication at a granular level. Aircrack-ng supports advanced attack techniques such as handshake capture and key cracking, while Reaver focuses on exploiting WPS vulnerabilities.

The study highlighted the effectiveness of these tools in professional cybersecurity environments, where accurate and detailed analysis is required. These tools are widely used by security professionals for vulnerability assessment, penetration testing, and network auditing. The ability to perform deep packet inspection and advanced analysis makes them highly reliable for identifying security flaws.However, the research also pointed out several limitations. These tools require high-performance systems and external Wi-Fi adapters that support monitor mode and packet injection. The setup process is often complex, making it difficult for beginners to use effectively. Additionally, the lack of portability limits their use in field-based scenarios.Overall, the study demonstrated that while traditional penetration testing tools are powerful, there is a need for simpler and more portable alternatives that can provide similar functionality with reduced complexity.

Limitations:

Despite their effectiveness, the use of traditional penetration testing tools introduces several practical limitations. One of the major constraints is the requirement for high-performance computing systems, as these tools typically rely on laptops or desktops with sufficient processing

power and memory. In addition, many of these tools require external Wi-Fi adapters that support monitor mode and packet injection, which increases the overall hardware dependency and cost.

The configuration and setup process for these tools is often complex and requires a strong understanding of networking concepts and command-line interfaces. This makes them less accessible to beginners or non-technical users, limiting their usability in educational environments. Furthermore, the lack of portability restricts their application in real-world or field-based scenarios, where compact and standalone solutions are more desirable.

Another limitation is that these tools are generally designed for specialized environments and may not integrate easily into lightweight or embedded systems. As a result, while they offer powerful capabilities, there is a need for simpler, portable, and integrated solutions that can provide similar functionalities with reduced complexity and improved accessibility.

2.3 Detection of Deauthentication Attacks in Wireless Networks

Author(s): S. Khan, 2022

Overview:

This study proposed a method for detecting deauthentication attacks in wireless networks by monitoring management frame activity. The system focuses on identifying abnormal increases in deauthentication frames, which are commonly used in denial-of-service attacks to disconnect clients from access points.The detection mechanism is based on a threshold-based approach, where the system continuously counts the number of deauthentication packets within a specific time interval. If the count exceeds a predefined threshold, the system identifies it as a potential attack. This approach is simple yet effective for detecting common wireless attacks.

The research demonstrated that such detection systems can operate efficiently in real time and require minimal computational resources. This makes them suitable for implementation on embedded platforms and IoT devices. The study also emphasized the importance of monitoring management frames, which are often overlooked in traditional security systems.However, the detection mechanism is limited to specific attack types and cannot identify more complex or multi-stage attacks. The system also lacks visualization tools for presenting detection results, making it less user-friendly.Despite these limitations, the study provides a practical approach for implementing lightweight intrusion detection systems in wireless networks.

Limitations:

While the proposed detection approach is effective for identifying deauthentication-based attacks, it is limited in scope and cannot detect more complex or multi-stage wireless attacks such as

EvilTwin or coordinated spoofing attacks. The system relies on a threshold-based mechanism,which may lead to false positives in high-traffic environments or false negatives when attackersoperate below detection thresholds. This dependence on predefined thresholds reduces adaptability to dynamic network conditions.Additionally, the model does not include attack simulation capabilities, which restricts its ability to validate vulnerabilities or evaluate system robustness under different attack scenarios. The absence of visualization tools or a user-friendly interface limits practical usability, particularly for non-technical users who require clear monitoring dashboards. Furthermore, the system lacks integration with advanced analytics or machine learning techniques, reducing its effectiveness in detecting evolving or previously unknown attack patterns. The approach also does not provide automated response mechanisms, making it primarily a passive detection system rather than an active defense solution.

2.4 Lightweight Intrusion Detection Systems for IoT Networks

Author(s): A. Mehmood et al., 2021

Overview:

This research focused on the development of lightweight intrusion detection systems specifically designed for IoT environments. The study addressed the challenges of implementing security mechanisms on devices with limited computational power, memory, and energy resources.

The proposed system monitors network traffic patterns and identifies anomalies using predefined rules and statistical analysis. It is designed to operate efficiently on embedded platforms, ensuring minimal impact on system performance. The research demonstrated that basic anomaly detection techniques can be effectively implemented on low-power devices.

The study emphasized the importance of scalability, where multiple IoT devices can be deployed to monitor network activity across different locations. This distributed approach enhances overall network visibility and improves detection coverage.

However, the system is limited in terms of detection capabilities and does not support advanced analysis techniques such as deep packet inspection or machine learning-based detection. It also lacks offensive capabilities, making it purely a defensive system.Overall, the research highlights the importance of developing efficient and lightweight security solutions for IoT environments.

Limitations:

Although the system is optimized for low-power and resource-constrained environments, it provides only basic detection capabilities and lacks support for advanced analysis techniques such as deep packet inspection, behavioral analysis, or machine learning-based detection. This limitation restricts its ability to identify sophisticated, polymorphic, or zero-day threats. The reliance on predefined rules and statistical thresholds reduces flexibility and may result in lower detection accuracy in

highly dynamic or complex network environments.The system is purely defensive in nature and does not support attack simulation or penetration testing features, which limits its applicability in comprehensive security evaluation frameworks. Additionally, the reduced computational overhead comes at the cost of limited feature richness, meaning that detailed traffic analysis and fine-grained inspection are not feasible. The system also lacks centralized management or visualization capabilities, which can make large-scale deployment and monitoring more challenging. Furthermore, scalability may introduce synchronization and coordination issues when multiple devices are deployed across distributed environments.

2.5 Analysis of Wi-Fi Sniffing and Packet Monitoring Techniques

Author(s): M. Conti, N. Dragoni, and V. Lesyk, 2016

Overview:

This study presented a comprehensive analysis of Wi-Fi sniffing and packet monitoring techniques, focusing on how network traffic can be captured, inspected, and analyzed to identify potential security threats. The research primarily examined the use of promiscuous mode in wirelessinterfaces, which allows a device to capture all packets transmitted over a wireless channel, regardless of their intended destination.The study explained that packet sniffing plays a crucial role in network monitoring, intrusion detection, and traffic analysis. By capturing raw wireless packets, the system can extract critical information such as source and destination addresses, frame types, protocol details, and communication patterns. This information is essential for understanding network behavior and identifying anomalies.

Furthermore, the research emphasized the importance of real-time packet analysis in detecting malicious activities such as spoofing attacks, deauthentication attacks, and unauthorized access attempts. By analyzing patterns in packet transmission, frequency, and structure, the system can identify deviations from normal behavior, which may indicate the presence of an attack.

The study also explored the implementation of packet sniffing techniques in lightweight and embedded environments. It demonstrated that embedded systems, including low-power devices, can be configured to perform continuous packet monitoring using optimized algorithms and efficient resource management. This makes packet sniffing a viable solution for portable and distributed network monitoring systems.However, the research highlighted several challenges associated with packet sniffing. One of the primary limitations is the complexity of analyzing raw packet data, which requires in-depth knowledge of networking protocols and packet structures. Additionally, processing large volumes of captured data can be computationally intensive, especially for resource-constrained devices.

The study also noted that packet sniffing alone isinsufficient for a complete security solution, as it focuses primarily on monitoring and does not include active attack simulation or response mechanisms.Overall, the study provides valuable insights into the role of packet sniffing in wireless security analysis and highlights its importance as a foundational technique for intrusion detection and network monitoring systems.

Limitations:

Despite its effectiveness in capturing and analyzing network traffic, packet sniffing techniques present several inherent challenges. One of the primary limitations is the complexity involved in processing raw packet data, which requires in-depth knowledge of networking protocols, frame structures, and wireless communication standards. This makes implementation and interpretation difficult for users without strong technical expertise.Additionally, processing large volumes of captured packets can introduce significant computational overhead and memory usage, particularly in high-density network environments. This can affect system performance, especially when implemented on resource-constrained embedded devices.

Packet sniffing techniques are also limited in functionality, as they focus primarily on passive monitoring and do not include active attack simulation, prevention, or automated response mechanisms.Furthermore, encrypted traffic limits the visibility of packet contents, reducing the effectiveness of analysis in modern secure networks. The approach also lacks contextual awareness, meaning it may identify anomalies but cannot always determine the intent or severity of an attack without additional processing layers. Finally, the dependency on expert analysis for accurate interpretation makes it less accessible for general users and limits its usability in automated security systems.

CHAPTER 3

EXISTING SYSTEM

3.1 Existing System

The existing approaches to Wi-Fi penetration testing and wireless threat analysis primarily rely on conventional software-based tools installed on laptops or penetration-testing workstations. These tools include frameworks such as Aircrack-ng, Wireshark, Kismet, Reaver and Hashcat, which are widely utilized in cybersecurity laboratories and ethical hacking environments. Although these platforms provide an extensive set of attack, monitoring and analysis capabilities, they require specialized hardware, operating system dependencies and technical expertise to operate effectively. In most cases, users must configure an external wireless adapter capable of monitor mode and packet injection, a requirement that introduces compatibility issues and additional hardware overhead.

These conventional systems operate within the host machine, meaning the user must already have a fully configured software environment before initiating any wireless audit. This creates a dependency chain involving operating system support, wireless chipset capability and terminal-based command execution, making the setup process complex for beginners. Additionally, Wi-Fi auditing and attack execution on such systems often follow a multi-tool workflow, requiring the user to manually move between scanners, packet analyzers, cracking engines and logging utilities. As a result, the process becomes fragmented, time-consuming and error-prone, especially for users lacking professional experience.

Furthermore, existing platforms focus heavily on offensive capabilities with minimal or no built-in defensive monitoring features. Tools like Aircrack-ng or ESP32 Marauder allow users to perform deauthentication attacks, handshake captures or rogue access point simulation, yet they do not provide real-time detection or alert mechanisms to help users understand the effects of these attacks or monitor live wireless anomalies. This creates a gap in practical education because learners can initiate attacks but do not receive immediate insight into how networks identify or respond to these threats.

3.2 Existing Methodology: Software-Based Wireless Penetration Tools

The traditional methodology for wireless penetration testing relies on a combination of scanning, packet capturing and cracking techniques performed through independent software utilities. When performing an assessment, the user typically begins by scanning surrounding wireless networks to identify access points, SSIDs, encryption types, channels and associated devices. Once a target is

selected, the tester may attempt to capture authentication frames such as WPA/WPA2 handshakes or PMKID data using tools like airodump-ng or Bettercap. The captured data is then exported to a cracking engine such as Hashcat or Aircrack-ng, which attempts to recover the password using dictionary-based or brute-force techniques.

This methodology is effective in controlled penetration-testing workflows; however, it remains resource-intensive and requires technical proficiency in command-line operations and wireless protocol structures. Additionally, the workflow is reactive and sequential, meaning the attacker conducts reconnaissance, prepares attacks and manually executes modules without automatic intelligence or integrated reporting. While these tools offer powerful capabilities for advanced analysts, they are not optimized for learners, trainers or institutions seeking a portable and structured learning platform. More importantly,existing methodologies lack integrated visualization interfaces. Although dashboards exist in frameworks such as Kali Linux or Wi-Fi Pineapple, they are either costly, proprietary or hardware-restricted. Therefore, learners frequently operate without intuitive feedback, making it difficult to interpret captured packets, identify anomalies or understand attack impact in real time.

3.3 Limitation Of The Existing System

Despite the widespread availability and technical maturity of current wireless penetration frameworks, they suffer from several architectural and operational limitations that restrict accessibility, portability and comprehensive learning. The most significant limitations are outlined below:

1.High Hardware and Setup Dependency: Current penetration testing platforms rely heavily on external Wi-Fi adapters capable of monitor mode and packet injection, along with Linux-based operating systems supporting specialized drivers. Configuring these components often requires technical expertise in wireless chipset compatibility, firmware patching, and operating system tuning, making setup time-consuming and error-prone. This dependency increases deployment cost and prevents spontaneous usage in environments outside a fully configured workstation. As a result, existing solutions remain inaccessible to individuals without advanced technical backgrounds or dedicated hardware resources.

2. Lack of Integrated Defensive Monitoring: Most existing Wi-Fi testing tools prioritize offensive capabilities, such as packet injection, handshake capture, and spoofing, rather than security monitoring or defensive analysis. They lack automated detection mechanisms to identify attacks like rogue access points, jamming, or deauthentication attempts while penetration testing operations are being executed. This creates a one-sided learning environmentwhere users simulate attacks without understanding real-time network responses or threat detection strategies.

The absence of unified offensive–defensive capability limits the educational value and fails to represent real-world cybersecurity environments where detection is as critical as attack execution.

3. Fragmented Workflow and Steep Learning Curve: Wireless auditing currently requires multiple tools executed independently for scanning, capturing, injection, cracking, and analysis, resulting in a fragmented workflow. Users must manually transfer results from one tool to another, requiring familiarity with command syntax, network protocols, and low-level packet structures. This creates a steep learning curve, especially for entry-level cybersecurity students or institutions attempting classroom-based training. Additionally, workflow complexity increases the likelihood of operational mistakes, reducing efficiency and discouraging adoption among non-technical or early-stage learners.

4. Limited Portability and Field Usability: Traditional penetration testing systems rely on laptops or desktop environments, making them unsuitable for lightweight field operations or mobile-based security assessments. These setups often require continuous power, storage capacity, and intact software configurations, which limits their practicality in environments like IoT testing labs, on-site audits, or rapid wireless investigations. Furthermore, the physical footprint and setup requirements reduce flexibility, preventing spontaneous vulnerability assessments or demonstrations in environments where time and mobility are restricted.

5. No Real-Time Visualization or Attack Feedback: Most existing tools execute wireless attacks without providing immediate visual feedback or analytical insight into the impact of those attacks. Users may perform deauthentication floods, spoofing, or handshake captures without knowing whether the network reacted, devices disconnected, or packet flow patterns changed. This lack of real-time responsiveness reduces the effectiveness of the learning experience and prevents clear understanding of wireless behaviour under attack conditions. Without visualization and contextual interpretation, cybersecurity training becomes theoretical rather than interactive or evidence-based.

CHAPTER 4

PROPOSED SYSTEM

4.1 Proposed System

The proposed system, titled Slipper Zero Plus: ESP32-Based Wi-Fi Penetration Testing and Intrusion Detection Toolkit, is designed to provide a comprehensive and portable solution for analyzing wireless network vulnerabilities while simultaneously detecting malicious or abnormal wireless activities in real time. Unlike traditional desktop-based penetration testing tools that depend on high-performance systems, external Wi-Fi adapters, and specialized operating environments, the proposed system integrates both offensive and defensive wireless security functionalities into a single embedded platform powered by the ESP32 microcontroller. This integration enables efficient wireless monitoring, attack simulation, and threat detection without requiring complex hardware configurations or external dependencies.

The system is developed as a practical platform for cybersecurity learning, research experimentation, and lightweight wireless auditing applications. It incorporates multiple offensive modules such as deauthentication attack simulation, beacon flooding, and rogue access point creation, which allow users to understand how wireless attacks are performed in real-world scenarios. In parallel, the system includes defensive mechanisms that analyze wireless traffic to detect abnormal behavior, such as excessive deauthentication frames, unusual packet patterns, and unauthorized device activity. This dual functionality ensures that users can observe both the execution and impact of wireless attacks as well as the corresponding detection techniques within a unified environment.By leveraging the native IEEE 802.11 capabilities of the ESP32 microcontroller, the system demonstrates that advanced wireless security operations can be implemented on low-cost embedded hardware.

The device operates in promiscuous mode, allowing it to capture all packets transmitted within its communication range. These packets are processed in real time to extract critical network information such as access point identifiers, connected client devices, signal strength, and frame types. This continuous packet monitoring provides a detailed understanding of network behavior and forms the foundation for both attack execution and intrusion detection.The architecture of the system is modular, consisting of distinct components responsible for scanning, attack execution, packet analysis, detection, and visualization. The scanning module continuously discovers nearby wireless networks and connected devices, while the attack module generates and injects specially crafted packets to simulate different attack scenarios.

The detection module analyzes captured packet data using threshold-based techniques to identify abnormal traffic patterns that may indicate ongoing attacks. These modules operate in coordination to ensure theefficient system performance and accurate analysis of wireless activities.In addition to its backend functionality, the system provides a web-based dashboard interface that enhances usability and accessibility. The dashboard is hosted directly on the ESP32 device and can be accessed through a standard web browser without requiring additional software installation. It provides real-time visualization of network data, including detected access points, connected devices, packet activity, and generated alerts. Users can interact with the system through intuitive controls to initiate scans, select targets, execute attack modules, and monitor detection results. This graphical interface simplifies the interaction process and makes the system suitable for both beginners and advanced users.

The proposed system also emphasizes portability and cost-effectiveness. By utilizing a single ESP32 module and open-source software components, the system eliminates the need for expensive hardware and complex setups. Its compact design allows it to be easily deployed in various environments, including classrooms, laboratories, and field testing scenarios. This makes it an accessible tool for students, researchers, and professionals interested in wireless security.Furthermore, the system is designed with scalability and flexibility in mind. The modular architecture allows new features, attack techniques, and detection algorithms to be integrated without requiring significant hardware modifications. This ensures that the system can adapt to evolving wireless security threats and technological advancements.

Overall, the proposed Slipper Zero Plus system provides a unified and efficient platform for wireless network analysis, attack simulation, and intrusion detection. By combining real-time packet monitoring, controlled attack execution, and intuitive visualization within a single embedded device, the system demonstrates that advanced Wi-Fi security operations can be achieved using low-cost hardware. This makes it highly suitable for cybersecurity education, practical experimentation, and lightweight wireless security assessment.

4.2 Objectives Of The Proposed System

The proposed system is designed with the following primary objectives:

• To develop a standalone embedded Wi-Fi penetration testing and intrusion detection system using the ESP32 platform that operates independently without requiring external laptops, additional wireless adapters, or specialized operating systems.

• To provide real-time wireless network monitoring by capturing packets using promiscuous mode and analyzing network activity to identify access points, connected clients, and communication patterns.

• To integrate offensive penetration testing features such as deauthentication attack simulation, beacon flooding, and rogue access point creation in order to demonstrate practical wireless attack scenarios in a controlled environment.

• To implement a lightweight intrusion detection mechanism capable of identifying abnormal network behavior, such as excessive deauthentication frames, unusual packet frequency, and suspicious wireless activity.

• To provide a web-based graphical dashboard that enables users to visualize network data, monitor real-time packet activity, control attack modules, and observe detection alerts through an intuitive interface.

• To ensure full portability and low power consumption by utilizing a compact ESP32-based embedded system suitable for field testing, academic laboratories, and cybersecurity training environments.

• To design a cost-effective and scalable platform using open-source technologies that can be extended with additional features such as advanced detection algorithms, support for new wireless protocols, and enhanced analytical capabilities.

4.3 Work Flow

The proposed system follows an automated and user-friendly workflow designed to provide a clear understanding of wireless security operations while ensuring efficient execution of scanning, attack simulation, and detection processes.

1. **Initialization:** Once the device is powered on, the ESP32 initializes all internal components including the Wi-Fi module, memory, and configuration settings. The system activates its Access Point mode, allowing users to connect to the device. Simultaneously, the web dashboard is initialized and becomes accessible through a browser, displaying system status and readiness for operation.
2. **Network Scanning:** The system enables promiscuous mode and begins scanning the surrounding wireless environment. It captures packets and extracts information such as SSID, MAC address, signal strength, channel, and encryption type of nearby access points. The collected data is continuously updated and displayed on the dashboard in real time.
3. **Selection and Action:** The user interacts with the dashboard interface to select a target network or choose a specific operation. Available actions include initiating deauthentication attacks, performing beacon flooding, creating a rogue access point, or continuing passive monitoring for detection purposes.
4. **Execution and Live Monitoring:** When an operation is selected, the system executes the corresponding function by generating and transmitting crafted packets or analyzing captured data.

During this phase, the dashboard provides live updates including packet activity, attack status, detected devices, and system logs. The detection module simultaneously monitors incoming packets to identify abnormal patterns.

5. **Result Display and Analysis:** The results of scanning, attack execution, and detection are displayed on the dashboard. Users can observe the impact of attacks, view detected anomalies, and analyze network behavior. Relevant data such as detected alerts, packet statistics, and device information are presented in a structured format for better understanding.
6. **Continuous Monitoring:** After completing an operation, the system returns to passive monitoring mode. It continues to capture and analyze wireless packets in real time, ensuring continuous detection of suspicious activities such as repeated deauthentication frames, rogue access points, or unusual traffic patterns.

4.4 Features Of The Proposed System

The Slipper Zero Plus system is designed to overcome the limitations of traditional Wi-Fi security tools by providing a compact, integrated, and user-friendly solution for wireless penetration testing and monitoring.

• **Standalone Operation:** The system operates entirely on the ESP32 microcontroller without requiring external computing devices, making it fully portable and self-contained for various deployment scenarios.

• **Real-Time Packet Monitoring:** Using promiscuous mode, the system continuously captures and analyzes wireless packets, providing real-time insights into network activity and communication patterns.

• **Offensive and Defensive Capabilities:** The system supports both attack simulation and intrusion detection, allowing users to perform penetration testing while simultaneously monitoring and identifying potential threats.

• **Attack Simulation Modules:** The system includes functionalities such as deauthentication attacks, beacon flooding, and rogue access point creation, enabling practical demonstration of common wireless vulnerabilities.

• **Lightweight Intrusion Detection:** A threshold-based detection mechanism is implemented to identify abnormal network behavior, such as high-frequency deauthentication packets or unusual traffic spikes.

• **Web-Based Dashboard Interface:** The system provides a browser-based graphical interface that allows users to control operations, monitor network activity, and view detection results without requiring command-line interaction.

• **Low-Cost and Open-Source Design:** Built using inexpensive hardware and open-source technologies, the system offers an affordable solution for students, researchers, and cybersecurity enthusiasts.

• **Modular and Extensible Architecture:** The system is designed with modular components, allowing easy integration of new features, attack techniques, and detection algorithms without hardware modification.

These features collectively make Slipper Zero Plus a flexible, efficient, and practical platform for studying wireless security concepts, performing controlled penetration testing, and analyzing network vulnerabilities in real-world environments.

4.5 Advantages Of The Proposed System

The proposed Slipper Zero Plus system offers several significant advantages over conventional wireless penetration testing tools and monitoring solutions. One of the primary advantages of the system is its lightweight, embedded, and portable design, which eliminates the dependency on high-performance computers, external wireless adapters, and specialized penetration testing environments. Traditional tools such as those used in desktop-based wireless auditing require complex configurations, driver support, and dedicated operating systems. In contrast, the proposed system integrates all functionalities within the ESP32 microcontroller and operates as a fully standalone unit. This portability allows the system to be easily deployed in various environments, including academic laboratories, field testing scenarios, and cybersecurity training sessions, without requiring additional infrastructure or setup complexity.

Another major advantage of the system is the integration of both offensive and defensive wireless security capabilities within a single unified framework. Existing solutions often focus exclusively on either penetration testing or network monitoring, requiring users to switch between multiple tools to understand the complete lifecycle of wireless attacks. The proposed system addresses this limitation by combining attack simulation techniques, such as deauthentication attacks, beacon flooding, and rogue access point creation, with real-time packet monitoring and intrusion detection mechanisms. This dual functionality enables users to observe how attacks are executed, analyze their impact on network behavior, and simultaneously detect abnormal activities. As a result, the system supports both offensive security testing and defensive analysis, making it highly valuable for comprehensive cybersecurity learning.The inclusion of a web-based dashboard interface further enhances the usability and accessibility of the system. Traditional wireless security tools often rely on command-line interfaces, scripting, and manual configuration, which can be challenging for beginners and time-consuming for experienced users.

The proposed system replaces these complexities with a browser-based graphical interface that provides intuitive controls, real-time data visualization, and structured output. Users can easily initiate scans, select target networks, execute attack modules, and monitor detection results without requiring advanced technical expertise. This significantly improves user interaction and makes the system suitable for a wide range of users, including students, educators, researchers, and cybersecurity professionals.

Cost efficiency is another important advantage of the proposed system. Conventional penetration testing setups often involve expensive hardware components, such as high-end laptops, specialized wireless adapters, and dedicated security platforms. In contrast, the Slipper Zero Plus system is built using a low-cost ESP32 microcontroller and open-source software components, making it an affordable solution for individuals and institutions. This cost-effective design enables wider adoption of wireless security tools and provides opportunities for hands-on learning without significant financial investment.

The system also benefits from its modular and extensible architecture, which allows future enhancements and customization without requiring major hardware modifications. Each functional component of the system, including scanning, attack execution, detection, and visualization, is designed as a separate module. This modular approach enables developers to add new features, integrate advanced detection algorithms, or support additional wireless protocols as technology evolves. The flexibility of the system ensures that it can adapt to emerging wireless security challenges and remain relevant in dynamic cybersecurity environments.

Overall, the proposed Slipper Zero Plus system addresses the limitations of traditional wireless penetration testing tools by offering a compact, cost-effective, and integrated solution that combines scanning, attack simulation, detection, and visualization within a single embedded platform. Its standalone operation, dual offensive and defensive functionality, user-friendly interface, and scalability make it a powerful tool for cybersecurity education, research, and practical wireless security assessment.

CHAPTER 5

SYSTEM REQUIREMENTS

5.1 Hardware Specification

The Slipper Zero Plus system is developed as a compact and embedded wireless security platform built around the ESP32 microcontroller, which serves as the core hardware component responsible for executing all wireless scanning, packet monitoring, attack simulation, and detection operations. The selected hardware architecture is designed to support real-time Wi-Fi packet capture using promiscuous mode as well as the transmission of crafted packets required for attack simulation. This enables the system to perform key security functions such as deauthentication attack execution, beacon frame generation, rogue access point creation, and continuous monitoring of wireless network activity.

The ESP32 microcontroller provides integrated Wi-Fi capabilities compliant with IEEE 802.11 standards, making it suitable for performing wireless operations without the need for external adapters. Its processing capability allows efficient handling of packet capture, analysis, and communication with the user interface. The compact and lightweight nature of the hardware ensures that the system remains portable and easy to deploy in various environments, including academic laboratories, field demonstrations, and cybersecurity awareness programs. This eliminates the dependency on bulky computing devices and enables practical experimentation using a single embedded unit.

The required hardware configuration is as follows:

• Microcontroller: ESP32 Development Board with integrated Wi-Fi module
• Processor: Dual-Core Xtensa LX6 processor operating up to 240 MHz
• Memory: Minimum 4 MB onboard flash memory
• Connectivity: IEEE 802.11 b/g/n wireless support with promiscuous mode capability
• User Access Device: Smartphone, laptop, or desktop system with web browser support
• Power Supply: 5V USB power source (USB cable, power bank, or adapter)
• Optional Components: External antenna for extended range, OLED display for standalone status visualization, protective enclosure casing

This hardware configuration ensures that the system is capable of performing real-time wireless monitoring, attack simulation, and detection while maintaining a low-cost and portable design suitable for embedded cybersecurity applications.

5.2 Software Specification

The software architecture of the Slipper Zero Plus system is designed as a layered framework consisting of embedded firmware, communication handling mechanisms, and a browser-based dashboard interface. Each layer is responsible for specific functionalities that collectively enable wireless scanning, attack execution, data processing, and real-time visualization.The embedded firmware running on the ESP32 is responsible for performing all core operations, including enabling promiscuous mode for packet capture, scanning nearby networks, executing attack modules, and analyzing packet data for intrusion detection. The firmware is developed using C/C++ in the Arduino IDE utilizing optimized Wi-Fi libraries, This enables the system to capture raw packets, process network information, and generate custom frames required for attack simulation.

The communication layer facilitates interaction between the ESP32 device and the user interface. The system operates in Access Point mode, allowing users to connect directly to the device and access the dashboard through a web browser. Data exchange occurs in real time, enabling continuous updates of network activity, packet information, and system status. This approach eliminates the need for external backend servers and simplifies the overall system architecture.The frontend dashboard provides an interactive interface for controlling system operations and visualizing results. It is implemented using web technologies such as HTML, CSS, and JavaScript, ensuring compatibility with standard browsers. The dashboard displays critical information including detected access points, connected devices, packet activity, and detection alerts.

The complete software stack includes:

- Firmware Platform: Arduino IDE or PlatformIO for ESP32 development.
- Programming Language: C/C++ for embedded firmware implementation.
- Wireless Libraries: ESP32 Wi-Fi libraries supporting promiscuous mode and packet manipulation.
- Frontend Technologies: HTML5, CSS3, JavaScript for dashboard interface.
- Visualization Tools: Dynamic UI components for displaying network data and alerts.
- Communication Mechanism: Web -based interface hosted on ESP32 (Access Point mode).

The combination of embedded firmware and web-based interface ensures efficient execution of wireless operations while maintaining simplicity and usability. The system is capable of handling real-time packet capture, attack simulation, and detection without requiring complex backend infrastructure.

CHAPTER 6

DESIGN AND IMPLEMENTATION

6.1 System Design

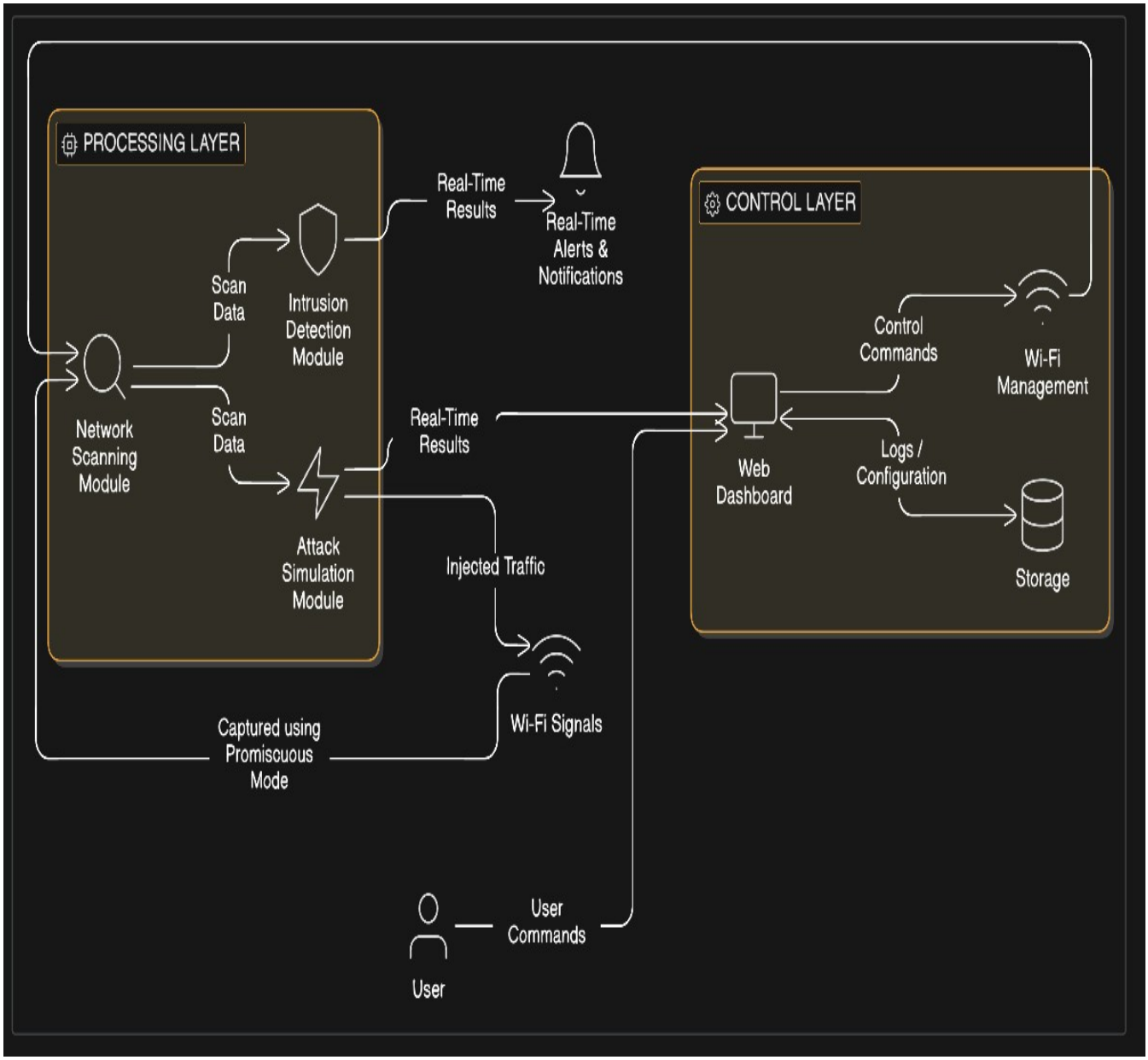


Fig 6.1.1: System Architecture

6.1.1 Architecture Design

The Slipper Zero Plus system is designed using a layered and modular architecture that clearly separates wireless data acquisition, processing, attack execution, and user interaction into distinct functional components.

This architectural approach ensures that the system remains scalable, maintainable, and adaptable to future enhancements such as advanced intrusion detection algorithms, AI-based threat analysis, and support for newer wireless protocols.In addition, the modular design enables independent development and testing of each component, reducing system complexity and improving overall reliability. It also allows easy integration of additional features without affecting existing functionalities, thereby supporting continuous system evolution.The architecture focuses on achieving real-time monitoring and control of wireless networks while enabling both offensive and defensive operations within the same framework.

By integrating these capabilities, the system provides a comprehensive platform for analyzing wireless security threats, understanding attack mechanisms, and demonstrating defense strategies. Furthermore, the system ensures efficient resource utilization on the ESP platform by balancing processing tasks and communication overhead, enabling smooth and stable operation even under high network activity

The system is divided into two major layers:

• Processing Layer

• Control Layer

These layers communicate continuously through real-time data exchange, ensuring synchronized operation between data processing and user interaction. This layered communication model enhances system responsiveness and ensures that user commands are executed promptly while maintaining continuous network monitoring.

6.2 Implementation

The implementation of the Slipper Zero Plus system is structured based on the two primary layers: the Processing Layer and the Control Layer. Each layer contains multiple modules that work together to achieve real-time wireless monitoring, attack simulation, and intrusion detection. The implementation follows a modular programming approach, where each functional component is developed as an independent unit and later integrated into the overall system. This approach improves code maintainability, simplifies debugging, and allows efficient updates or enhancements in future versions of the system.

6.2.1 Processing Layer

The Processing Layer serves as the core operational unit of the system, responsible for capturing wireless signals, analyzing network behavior, and executing controlled attack simulations. It directly interacts with the Wi-Fi environment and performs all computation-intensive tasks. This layer operates in real time, continuously processing incoming wireless packets and responding to dynamic changes in the network environment. It ensures that all critical operations such as packet capture, data analysis, and attack execution are performed efficiently without introducing the

significant latency.The Processing Layer is designed to handle both offensive and defensive functionalities simultaneously.

1. Network Scanning Module

The Network Scanning Module is responsible for continuously monitoring the wireless environment and capturing network data using promiscuous mode. In this mode, the ESP32 device is capable of receiving all wireless packets transmitted within its communication range, regardless of their intended destination. This capability allows the system to passively observe the behavior of nearby wireless networks without actively associating with them. As a result, the module provides a comprehensive view of the surrounding Wi-Fi environment, which is essential for both analysis and security evaluation.

The module performs detailed network discovery by identifying all available access points within range and extracting key attributes associated with each network. These attributes include the Service Set Identifier (SSID), Basic Service Set Identifier (BSSID), operating channel, and signal strength measured in terms of Received Signal Strength Indicator (RSSI). By continuously capturing and updating this information, the system is able to maintain an accurate and dynamic representation of the wireless landscape. In addition to access point detection, the module also identifies connected client devices by analyzing transmitted frames and extracting their MAC addresses, thereby providing insights into network usage and connectivity patterns.

Another important function of the Network Scanning Module is the monitoring of beacon frames and probe requests. Beacon frames are periodically broadcast by access points to announce their presence, while probe requests are sent by client devices searching for available networks. By analyzing these frames, the system can determine network activity levels, detect hidden networks, and understand how devices interact within the wireless environment. This information is crucial for identifying abnormal behavior and potential security threats.

The captured packet data is further processed within the module to extract only relevant information required for subsequent operations. This preprocessing step involves filtering unnecessary data, organizing packet details into structured formats, and preparing the information for efficient analysis. The processed data is then forwarded to other system components, such as the Intrusion Detection Module and the Attack Simulation Module, enabling them to perform their respective functions based on accurate and up-to-date network information.In addition, the Network Scanning Module supports continuous and periodic scanning cycles to ensure that transient networks and devices are not overlooked. Wireless environments are highly dynamic, with devices frequently connecting and disconnecting.By performing repeated scans at defined intervals, the system is able to capture these changes and maintain an updated view of network conditions

This continuous monitoring capability enhances the reliability of the system and ensures that all relevant network activity is accurately recorded and analyzed.

2. Intrusion Detection Module

The Intrusion Detection Module is responsible for monitoring wireless network activity and identifying abnormal patterns that may indicate potential security threats. This module plays a critical role in the system by providing defensive capabilities that complement the offensive features of the Attack Simulation Module. It operates continuously, analyzing network traffic in real time and ensuring that any suspicious behavior is detected and reported promptly.

The detection process begins with the analysis of packets captured by the Network Scanning Module. The system focuses particularly on management frames such as deauthentication frames and beacon frames, which are commonly used in wireless attacks. By monitoring the frequency and distribution of these frames, the module can identify unusual activity that deviates from normal network behavior. This allows the system to detect ongoing attacks such as deauthentication attacks and beacon flooding with a reasonable level of accuracy.

The module primarily uses a threshold-based detection mechanism, where predefined limits are set for specific types of network activity. For example, if the number of deauthentication packets exceeds a certain threshold within a given time period, the system interprets this as a potential attack. Similarly, an excessive number of beacon frames may indicate a beacon flooding scenario. This rule-based approach provides a simple yet effective method for detecting common wireless threats.

In addition to threshold-based detection, the module also considers temporal patterns and behavioral trends within the network. By analyzing how network parameters change over time, such as fluctuations in signal strength, packet transmission rates, and device activity, the system can identify anomalies that may not be immediately apparent through static rules. This enhances the reliability of the detection process and reduces the likelihood of false positives.

When suspicious activity is detected, the module generates alerts and logs detailed information about the event. These alerts include details such as the type of detected threat, associated devices, timestamp, and relevant network parameters. The logged data is then forwarded to the Control Layer, where it is displayed on the user dashboard. This enables users to monitor network activity in real time and take appropriate action when necessary.The module also supports continuous monitoring, ensuring that the network is observed at all times without interruption.This is particularly important in dynamic wireless environments where conditions can change rapidly.By maintaining constant surveillance, the system is able to detect both short-term and long-term anomalies, providing a comprehensive view of network behavior.

Furthermore, the Intrusion Detection Module works in coordination with other system components to provide a unified security framework. It receives processed data from the scanning module and correlates it with attack activity generated by the Attack Simulation Module. This integration allows the system to validate detection mechanisms and improve overall accuracy by comparing expected and observed behavior.In conclusion, the Intrusion Detection Module enhances the system by providing real-time monitoring and threat detection capabilities. It enables the identification of abnormal wireless activity and supports informed decision-making through detailed alerts and logs. By combining detection with attack simulation, the system achieves a balanced approach to wireless security, making it an effective tool for both analysis and research purposes.

3. Attack Simulation Module

The Attack Simulation Module is responsible for generating controlled wireless attack scenarios in order to analyze network vulnerabilities and demonstrate potential security threats in a structured and observable manner. This module operates using the information obtained from the Network Scanning Module, which provides details about available access points, connected client devices, and network characteristics. By leveraging this information, the system is able to identify appropriate targets and execute attack simulations in a controlled and selective way, ensuring that the testing process remains precise and relevant to the current wireless environment.

The module implements several well-known wireless attack techniques, each designed to replicate real-world threat scenarios. One of the primary techniques used is the deauthentication attack, where the system transmits forged deauthentication frames to disconnect client devices from legitimate access points. This type of attack exploits the lack of authentication in management frames within certain Wi-Fi standards. Another technique implemented is beacon flooding, where the system generates and broadcasts a large number of fake access points with different SSIDs. This creates network congestion and can confuse users by populating their device with multiple illegitimate network options.In addition to these techniques, the module also supports the implementation of Evil Twin attacks. In this scenario, the system creates a rogue access point that mimics a legitimate network by copying its SSID and configuration parameters. When users connect to this fake network, they may be redirected to a captive portal interface where sensitive information such as login credentials can be collected. This technique demonstrates the risks associated with unsecured wireless environments and highlights the importance of user awareness in network security.The Attack Simulation Module is designed with configurable parameters that allow users to control the behavior and intensity of the attacks.

Parameters such as packet transmission rate, attack duration, target selection, and broadcast intervals can be adjusted according to the testing requirements. This flexibility enables users to simulate different attack scenarios and observe their impact on network performance and client

devices. The system ensures that packet transmission is optimized to maintain stability while still achieving effective simulation results.

To ensure responsible and ethical usage, the module is intended to operate within controlled environments such as laboratories or authorized testing setups. The execution of attacks is coordinated with other modules in the system to allow simultaneous monitoring and analysis. All attack activities are recorded, including details such as target devices, number of packets transmitted, and duration of the attack. These logs provide valuable data for evaluating system performance and understanding the effects of different attack strategies.

Furthermore, the module is designed to work seamlessly with the Intrusion Detection Module, enabling real-time observation of attack behavior and detection responses. This integration allows users to study both offensive and defensive aspects of wireless security within a single framework. By observing how attacks are detected and handled, users can gain a deeper understanding of security mechanisms and improve defensive strategies.

Overall, the Attack Simulation Module provides a comprehensive platform for studying wireless attack techniques in a practical and controlled manner. It enables users to replicate real-world threat scenarios, analyze their impact, and evaluate system responses. This contributes significantly to the overall objective of the system, which is to provide a balanced and integrated approach to wireless security analysis.

6.2.2 Control Layer

The Control Layer functions as the primary interface between the user and the system, enabling effective interaction, monitoring, and control of all system operations. This layer is responsible for managing user inputs, presenting processed data in a structured format, and coordinating the execution of various system functionalities. It acts as a bridge that connects the user-facing components with the underlying processing mechanisms, ensuring that commands are accurately interpreted and executed. By abstracting the complexity of wireless operations, the Control Layer enhances usability and allows users to interact with the system without requiring in-depth technical knowledge.The Control Layer is designed to provide real-time feedback and control capabilities, ensuring that users can observe system behavior and respond to events as they occur. It integrates multiple components that collectively handle visualization, communication, and data management.Through continuous data exchange with the Processing Layer, the Control Layer ensures synchronization between system operations and user interaction. This coordinated approach allows the system to maintain responsiveness while processing complex wireless data in real time.

Additionally, the Control Layer ensures efficient error handling and system stability by managing unexpected conditions and maintaining smooth operation across all modules.

It also facilitates scalability by allowing seamless integration of new features and modules without affecting existing system functionality.

1. Web Dashboard

The Web Dashboard serves as the primary user interface of the system, providing a centralized platform for monitoring, analysis, and control. It is designed to present complex wireless network data in an intuitive and user-friendly format, enabling users to easily understand system behavior and network conditions. The dashboard displays real-time information about detected access points, connected client devices, signal strength, and network parameters, allowing users to gain a comprehensive view of the wireless environment.

The interface also visualizes system performance indicators and packet activity, helping users analyze network behavior and identify anomalies. Alerts and notifications generated by the Intrusion Detection Module are prominently displayed, ensuring that users are immediately informed of any suspicious activity. These alerts provide essential details such as the type of detected threat, associated devices, and time of occurrence, enabling users to make informed decisions regarding system actions.

In addition to monitoring capabilities, the dashboard provides interactive controls that allow users to initiate scanning operations, select target networks, and execute attack modules. These controls are integrated into the interface in a structured manner, ensuring ease of use and accessibility. Users can dynamically interact with the system without requiring command-line inputs, which significantly improves usability for beginners and non-technical users.

The dashboard operates using dynamic data updates, ensuring that information is continuously refreshed without requiring manual intervention. This real-time update mechanism allows users to observe changes in network conditions and system behavior as they occur. Data is organized into tables, charts, and visual elements that enhance readability and interpretation. This structured presentation enables users to quickly identify important information and analyze trends within the wireless environment.

Furthermore, the Web Dashboard plays a critical role in improving the overall user experience by providing a cohesive and interactive interface. It simplifies the complexity of wireless security analysis and allows users to focus on understanding network behavior rather than managing system operations. As a result, the dashboard significantly enhances the accessibility and effectiveness of the system.Additionally, the dashboard can incorporate customizable filters and sorting options, allowing users to focus on specific networks, devices, or threat levels for more efficient analysis. It can also support responsive design and cross-device compatibility, enabling seamless access and monitoring from desktops, tablets, and mobile devices. It can further include role-based access control to ensure secure and restricted usage of system functionalities among different users.

2. Wi-Fi Management Module

The Wi-Fi Management Module is responsible for handling communication between the user interface and the underlying hardware and processing components. It acts as a control bridge that ensures user commands are accurately transmitted to the appropriate modules within the system. This module plays a crucial role in coordinating system operations and maintaining synchronization between different functional components.

One of the primary responsibilities of this module is to receive user commands from the Web Dashboard and forward them to the Processing Layer. These commands may include initiating network scans, selecting targets, or executing attack simulations. The module ensures that each command is interpreted correctly and executed in the appropriate sequence, preventing conflicts between simultaneous operations.

In addition to command handling, the module manages scanning operations and controls the switching between different operational modes, such as monitoring mode and attack mode. This dynamic mode management allows the system to adapt to different use cases and ensures that the correct functionality is activated based on user requirements. The module also regulates wireless transmission parameters, including packet transmission rate and channel selection, to maintain system stability and performance.

The Wi-Fi Management Module is also responsible for handling communication protocols required for transmitting and receiving wireless signals. It ensures proper interaction with the ESP32 hardware, enabling efficient packet capture and transmission. By maintaining synchronization between hardware operations and software processes, the module ensures smooth and reliable system performance.Furthermore, the module plays a key role in coordinating the activities of different system components. It ensures that scanning, detection, and attack processes operate in a synchronized manner without interfering with each other. This coordination is essential for maintaining system efficiency and ensuring accurate results during operation.

Overall, the Wi-Fi Management Module acts as the central control unit of the system, enabling seamless communication between the user interface and the processing components. Its ability to manage commands, control operations, and maintain synchronization significantly contributes to the overall functionality and reliability of the system.

3. Storage Module

The Storage Module is responsible for maintaining persistent records of system activity, configurations, and analysis results. It plays a vital role in enabling data tracking, historical analysis, and reporting. By storing detailed information about system operations, the module allows users to review past events and gain insights into network behavior over time.

The module stores various types of data, including historical scan results, which contain information about detected networks, access points, and connected client devices. This data provides a record of network activity and can be used to analyze changes in the wireless environment. Additionally, the module records security alerts generated by the Intrusion Detection Module, along with timestamps and associated details, enabling users to track potential threats.

Another important function of the Storage Module is maintaining logs of attack execution. These logs include information about the type of attack performed, target devices, duration, and packet transmission statistics. This data is valuable for evaluating the effectiveness of attack simulations and understanding their impact on network behavior.

The module also stores user-defined configurations and system settings, allowing the system to retain preferences and operational parameters across sessions. This ensures consistency in system behavior and reduces the need for repeated configuration. The stored data can be accessed and analyzed to generate reports, support research activities, and facilitate learning.

Furthermore, the Storage Module enhances system transparency and accountability by maintaining comprehensive records of all activities. These records can be used for auditing purposes and for understanding the sequence of events during system operation.By providing persistent data storage, the module supports long-term analysis and contributes to the overall effectiveness of the system. The module can also implement data compression and efficient indexing techniques to optimize storage usage and improve retrieval speed for large datasets.Additionally, secure storage mechanisms such as encryption and access control can be integrated to protect sensitive logs and ensure data integrity.

6.2.3 Data Flow and Interaction

The interaction between the Processing Layer and the Control Layer forms a continuous operational cycle that enables real-time monitoring, analysis, and control of wireless networks. This interaction ensures that data captured from the wireless environment is processed, analyzed, and presented to the user in a seamless and synchronized manner. The system operates as an integrated framework where each component contributes to the overall functionality.

The process begins with the Network Scanning Module capturing wireless signals using promiscuous mode. The captured data is then processed and forwarded to both the Intrusion Detection Module and the Attack Simulation Module for further operations. The Intrusion Detection Module analyzes the data to identify abnormal patterns and generates alerts when suspicious activity is detected. At the same time, the Attack Simulation Module executes user-defined attack commands by injecting crafted packets into the network.

The results generated by these modules are transmitted to the Control Layer, where they are processed and displayed on the Web Dashboard.The dashboard provides real-time updates,allowing users to monitor network activity, view alerts, and analyze system performance. This continuous feedback mechanism ensures that users are always informed about the current state of the system.User inputs provided through the dashboard are sent back to the system via the Wi-Fi Management Module, which ensures proper execution of commands. This creates a feedback loop where user actions influence system behavior, and system outputs are continuously updated based on these actions. This bidirectional communication is essential for maintaining real-time interaction.

All system activities, including scanning results, alerts, and attack logs, are recorded by the Storage Module. This ensures that data is preserved for future reference and analysis. The stored information can be used to evaluate system performance, identify trends, and support further research.Overall, the data flow mechanism ensures smooth coordination between all system components, enabling efficient operation and real-time responsiveness. This continuous cycle of data capture, processing, analysis, and user interaction forms the foundation of the system's functionality and effectiveness.

6.3 Hardware Circuit Design

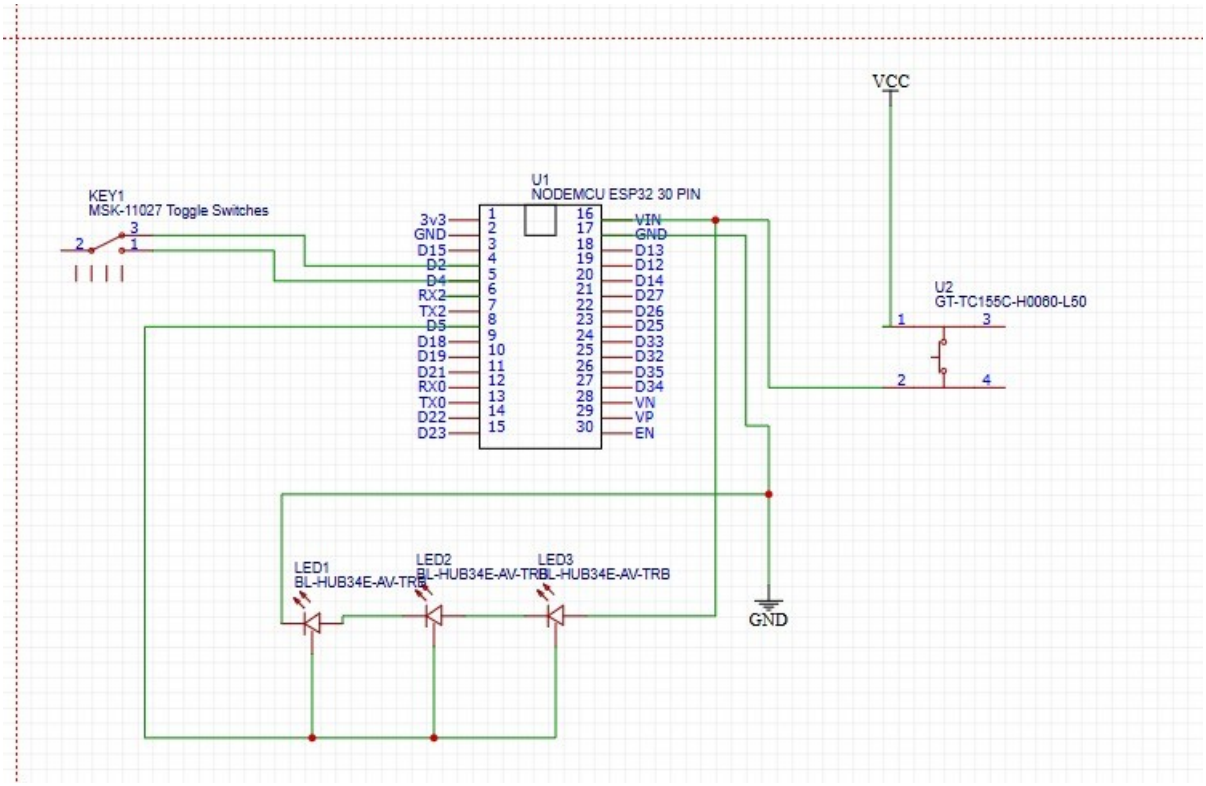


Fig 6.3.1 :Circuit Diagram of ESP32-Based Wi-Fi Security System

The above figure represents the hardware-level circuit design of the proposed ESP32-based wireless security analysis system.

The circuit is primarily centered around the NodeMCU ESP32development board, which acts as the core processing unit responsible for executing all system operations.These operations include wireless network scanning, packet capture in promiscuous mode, intrusion detection, and controlled attack simulation. The selection of the ESP32 platform is based on its integrated Wi-Fi capabilities, high processing efficiency, and support for advanced networking features, making it suitable for implementing both offensive and defensive wireless security mechanisms within a compact embedded system.The ESP32 module plays a critical role in enabling direct interaction with the wireless environment. Its ability to operate in promiscuous mode allows it to capture all nearby Wi-Fi packets without requiring association with any specific network. Additionally, the support for raw packet injection enables the system to simulate various wireless attacks such as deauthentication and beacon flooding. These capabilities eliminate the need for external network adapters or high-performance computing devices, thereby simplifying the overall system design and reducing hardware dependencies.

The circuit design incorporates a set of input switches connected to the General Purpose Input/Output (GPIO) pins of the ESP32. These switches serve as hardware-based control inputs that allow users to manually trigger specific system functions. For instance, users can initiate scanning operations, activate attack modes, or reset the system using these switches. The inclusion of physical input controls enhances system flexibility and ensures that basic operations can be performed even without access to the web-based dashboard. This feature is particularly useful in field deployments where direct interaction with the interface may not be feasible.

In addition to input controls, the circuit includes multiple LED indicators that provide real-time visual feedback regarding system activity. Each LED is assigned to represent a specific operational state, such as power status, active scanning, or attack execution. These indicators help users quickly identify the current state of the system without needing to monitor the dashboard continuously. The use of visual indicators improves usability and provides an immediate understanding of system behavior, which is especially beneficial during testing and demonstration scenarios.

The power supply section of the circuit is designed to ensure stable and reliable operation of the ESP32 and associated components. The system is powered using a regulated 5V USB supply, which is internally stepped down to the required voltage levels suitable for the ESP32. Proper voltage regulation is essential to prevent fluctuations that may affect system performance. Additionally, appropriate grounding connections are maintained throughout the circuit to minimize electrical noise and ensure signal integrity. This contributes to the overall stability and reliability of the hardware setup.The circuit also includes carefully designed interconnections between various components to facilitate seamless communication and control. GPIO pins are assigned based on functional requirements, ensuring that input and output operations do not conflict with each other.

This careful allocation of pins is important for maintaining compatibility with the firmware and ensuring efficient execution of system operations. The wiring layout is kept simple and organized to reduce complexity and improve maintainability.Furthermore, the hardware design emphasizes portability and compactness, allowing the system to be easily deployed in different environments. The use of minimal external components reduces the overall size and complexity of the circuit, making it suitable for mobile or field-based applications. This portability enables users to perform wireless security analysis in real-world scenarios, enhancing the practical applicability of the system.

Overall, the hardware circuit design complements the software architecture by providing a stable, efficient, and responsive platform for executing wireless security operations. The integration of input controls, visual indicators, and reliable power management ensures smooth system functionality. By combining simplicity with advanced capabilities, the hardware design supports the overall objective of developing a cost-effective and portable wireless security analysis system.

6.4 System Workflow

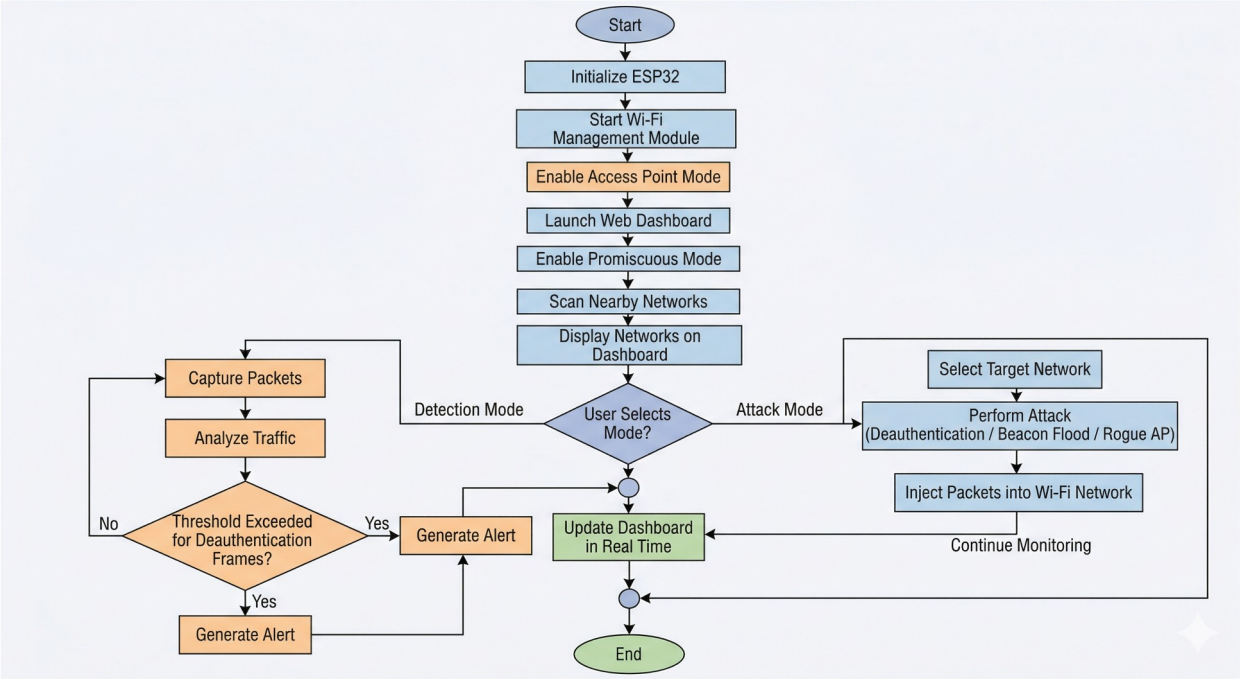


Fig 6.4.1 :System Flow Diagram

After initialization, the system enables Access Point (AP) mode and launches the web-based dashboard interface.This allows users to connect to the ESP32 device through a web browser and interact with the system in real time. The dashboard serves as the primary interface for controlling system operations, viewing detected networks, and monitoring system performance.

Following this, the device switches to promiscuous mode, which enables it to capture all wireless packets transmitted within its range, regardless of their destination. This capability is essential for performing detailed network analysis and monitoring wireless activity.Once promiscuous mode is activated, the system begins scanning the surrounding wireless environment. During this phase, the Network Scanning Module identifies available access points, extracts relevant network parameters, and detects connected client devices. The collected data is processed and displayed on the dashboard, providing users with a real-time view of the wireless landscape. This stage forms the foundation for subsequent operations, as it provides the necessary information required for both detection and attack functionalities.

At this point, the user selects the desired mode of operation, which can be either detection mode or attack mode. In detection mode, the system continuously captures and analyzes network traffic to identify abnormal patterns. The Intrusion Detection Module evaluates parameters such as packet frequency and frame types, particularly focusing on deauthentication frames and beacon activity. If the observed activity exceeds predefined thresholds, the system classifies it as suspicious and generates alerts. These alerts are displayed on the dashboard and may also be indicated through hardware components such as LEDs, ensuring immediate user awareness.

In attack mode, the system enables controlled execution of wireless attack simulations based on user input. The user selects a target network and chooses the type of attack to be performed, such as deauthentication, beacon flooding, or Evil Twin attack. The Attack Simulation Module then generates and injects crafted packets into the wireless network. During this process, the system continuously updates the dashboard with real-time information regarding attack status, packet transmission count, and target details. This allows users to observe the impact of the attack and analyze network behavior under simulated threat conditions.

Following the execution of either detection or attack operations, the system continues to monitor network activity in real time. This continuous monitoring ensures that any changes in network behavior or new threats are detected promptly. The system maintains an ongoing cycle of scanning, analysis, execution, and feedback, allowing it to adapt dynamically to the wireless environment. All relevant data, including scan results, alerts, and attack logs, are recorded for future reference and analysis.

Overall, the system workflow demonstrates a well-structured and automated approach to wireless network analysis and penetration testing. By integrating initialization, scanning, detection, attack execution, and continuous monitoring into a unified process, the system ensures efficient operation and real-time responsiveness. This workflow not only enhances system functionality but also provides a clear and systematic method for understanding and evaluating wireless security threats.

6.5 Code Implementation

```
#include "Scan.h"
#include "Attack.h"
#include "wifi.h"

Scan scan;
Attack attack;

void setup() {
    Serial.begin(115200);

    // Initialize Wi-Fi module
    wifi::begin();

    // Enable promiscuous mode for packet capture
    wifi_set_promiscuous_rx_cb([](uint8_t* buf, uint16_t len) {
        scan.sniffer(buf, len);
    });

    // Initialize scanning and attack modules
    scan.setup();

    // Start Access Point for Web Dashboard
    wifi::startAP();
}
```

6.5.1 System Initialization Code

This code initializes the ESP32 system, configures Wi-Fi functionality, and enables promiscuous mode for capturing wireless packets. The callback function forwards captured packets to the scanning module for further analysis. The system also starts an Access Point to host the web dashboard interface.

```
void loop() {
    // Update Wi-Fi operations
    wifi::update();

    // Perform network scanning
    scan.update();

    // Execute attack operations
    attack.update();
}
```

6.5.2 Main Execution Loop

The main execution loop defines the continuous operational cycle of the system. Within this loop, the system repeatedly updates Wi-Fi operations, performs network scanning, and executes attack-related processes. This ensures that all modules operate simultaneously in real time without interruption. The loop structure enables the system to dynamically respond to changes in the wireless environment, maintain continuous monitoring, and process user commands efficiently through the control interface.

```
wifi_set_promiscuous_rx_cb([](uint8_t* buf, uint16_t len) {
    scan.sniffer(buf, len);
});
```

6.5.3 Promiscuous Mode Packet Capture

This code implements promiscuous mode packet capture, which allows the ESP32 to receive all wireless packets within its range regardless of their intended destination. The callback function is triggered for every captured packet and forwards the raw data to the sniffer function in the scanning module. This mechanism forms the foundation for network monitoring and intrusion detection, as it enables real-time analysis of wireless traffic at the packet level.

```
#ifndef AP_SSID
#define AP_SSID "slipper_zero"
#endif

#ifndef AP_PASSWD
#define AP_PASSWD "espwifiex"
#endif
```

6.5.4 Access Point Configuration

This configuration defines the credentials for the ESP32's Access Point, including the SSID and password. By operating as a standalone Wi-Fi hotspot, the ESP32 allows users to connect directly to the system without requiring an external network. This enables the hosting of a web-based dashboard through which users can monitor network activity, initiate scans, and control system operations in a self-contained environment.

```
#define DEAUTHS_PER_TARGET 25
#define ATTACK_TIMEOUT 600
#define ATTACK_ALL_CH false
```

6.5.5 Attack Configuration Parameters

This section defines key parameters that control the behavior of attack simulations. It includes settings such as the number of deauthentication frames sent per target, the duration of attack execution, and whether the attack should operate across multiple channels. These parameters ensure that attack operations are executed in a controlled and configurable manner, allowing users to simulate different wireless attack scenarios while maintaining system stability.

```
// Global counters
volatile int deauthCount = 0;
unsigned long lastCheckTime = 0;

#define DEAUTH_THRESHOLD 5
#define TIME_WINDOW 5000   // 5 seconds
```

6.5.7 IDS Logic

This code initializes the core parameters required for the Intrusion Detection System. It defines counters and timing variables used to monitor suspicious network activity over a specified time window. The system tracks the number of detected deauthentication frames and compares it against a predefined threshold to identify abnormal behavior. This approach enables the system to detect potential wireless attacks based on traffic patterns rather than isolated events.

```
void scan::sniffer(uint8_t* buf, uint16_t len) {
    if (len < 24) return;

    uint8_t frameType = buf[0];

    // Deauthentication frame check (0xC0)
    if (frameType == 0xC0) {
        deauthCount++;
    }
}
```

6.5.8 Sniffer-based Detection

This implementation illustrates how the system performs packet-level analysis to detect potential attacks. The sniffer function processes captured packets, extracts the frame type, and identifies deauthentication frames based on their protocol value. Each detected frame increments a counter, allowing the system to measure the frequency of such events. This method provides a practical and protocol-based approach to intrusion detection, enabling real-time identification of suspicious wireless activity

```
#include <WiFi.h>
#include <WebServer.h>

WebServer server(80);

// Route: Get Scan Results
void handleScan() {
    String response = scan.getResults();
    server.send(200, "application/json", response);
}

// Route: Get Attack Status
void handleAttack() {
    String response = attack.getStatus();
    server.send(200, "application/json", response);
}

void setupServer() {
    // Define API endpoints
    server.on("/scan", HTTP_GET, handleScan);
    server.on("/attack", HTTP_GET, handleAttack);

    // Start server
    server.begin();
}
```

6.5.9 Web Server Initialization

This figure illustrates the implementation of the web server on the ESP32 using the `WebServer` library. The server is initialized on port 80 and defines API endpoints to handle client requests. The `/scan` endpoint returns real-time network scanning results, while the `/attack` endpoint provides the current attack status. The server enables communication between the ESP32 device and the web-based dashboard, allowing users to monitor and control system operations remotely.

CHAPTER 7

RESULT

The proposed Slipper Zero Plus Wi-Fi Penetration and Defence Toolkit was successfully implemented and evaluated using an ESP32-based embedded platform integrated with a backend server and a web-based dashboard interface. The system was tested in a real wireless environment to analyze its capability in detecting networks, monitoring devices, executing controlled attacks, and capturing sensitive authentication data through simulated attack scenarios.The experimental results confirm that the system effectively performs both offensive and defensive operations in real time. The ESP32 module successfully operates in promiscuous mode to capture wireless packets, while the dashboard provides a structured and interactive visualization of network activities. The integration of scanning, attack execution, and monitoring modules enables the system to function as a complete wireless security analysis platform.

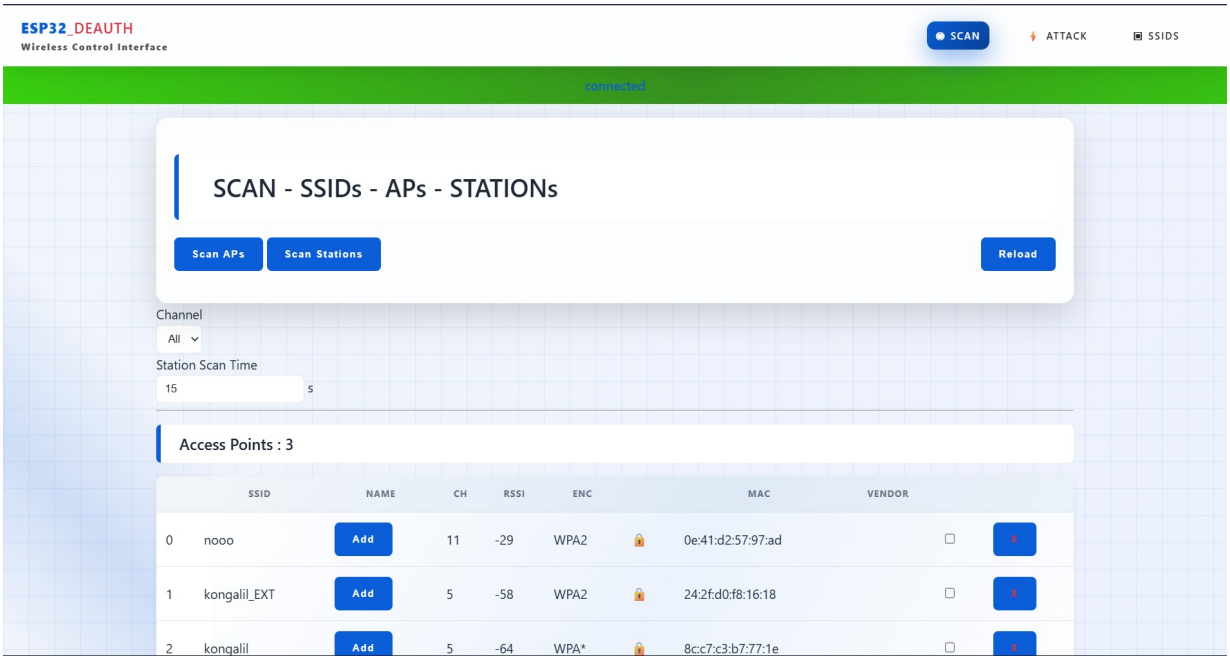


Fig 7.1: Network Scanning Interface

The above figure shows the scanning interface of the system, where the ESP32 performs wireless network discovery and displays detected access points. The interface includes options such as “Scan APs” and “Scan Stations,” allowing users to initiate scanning operations dynamically.During testing, the system successfully detected multiple access points and displayed key parameters including SSID, channel number, signal strength (RSSI), encryption type, MAC address, and vendor information. The system identified networks operating on different channels with varying signal strengths, indicating proper packet capture and parsing functionality.

The encryption field correctly distinguished between WPA2-secured networks and weaker configurations such as WPA. The RSSI values provided insight into signal strength and proximity of access points. The system also allowed users to select specific targets for further analysis or attack execution.This result confirms that the Network Scanning Module is capable of accurately detecting wireless networks and presenting structured information in real time.

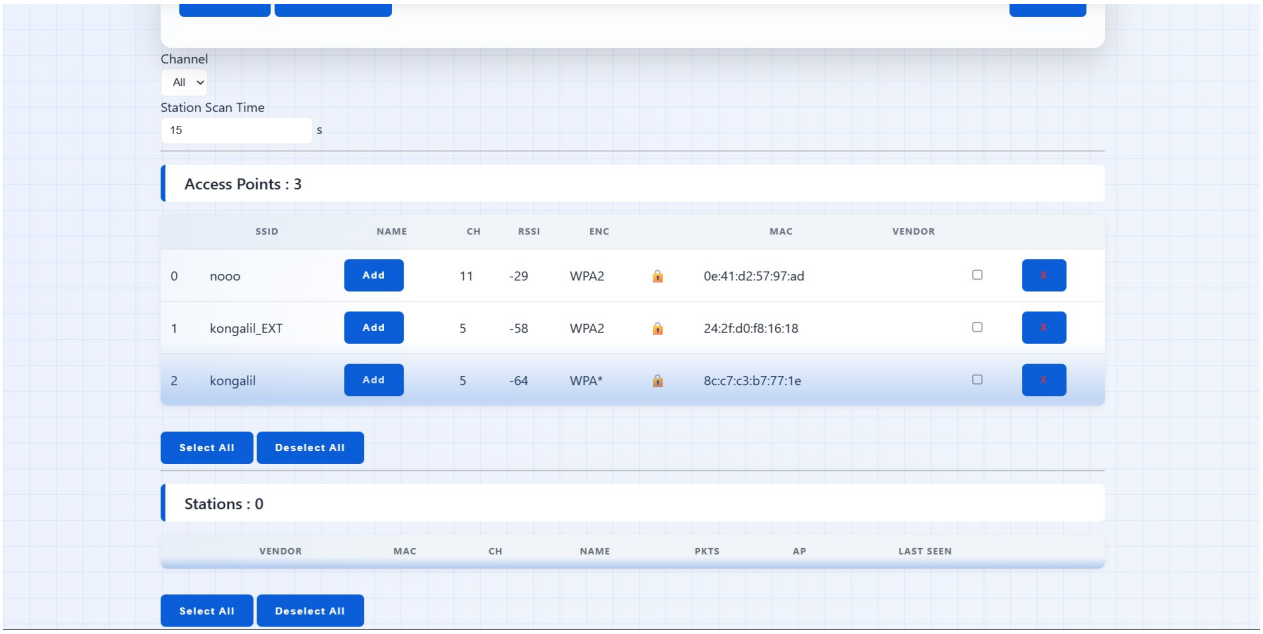


Fig 7.2: Access Point Enumeration and Selection

The above figure illustrates the detailed listing of detected access points along with user interaction features. The system displays multiple access points simultaneously and allows users to select or deselect targets using control buttons. The interface includes functionalities such as “Select All” and “Deselect All,” enabling efficient management of multiple targets. Each access point entry includes detailed attributes such as SSID, channel, RSSI, encryption type, and MAC address.In addition to basic listing, the system dynamically updates the access point information in real time as new networks are detected or existing ones change their parameters. This ensures that the user always has access to the most current network state.

The interface also supports filtering and sorting based on parameters such as signal strength or channel, allowing users to prioritize specific targets during analysis.During testing, the system successfully maintained a list of detected access points and allowed dynamic interaction without affecting system performance. The ability to select targets directly from the dashboard simplifies the process of initiating attacks or further monitoring. The system also ensures that user selections are synchronized with backend processing, enabling accurate execution of commands.

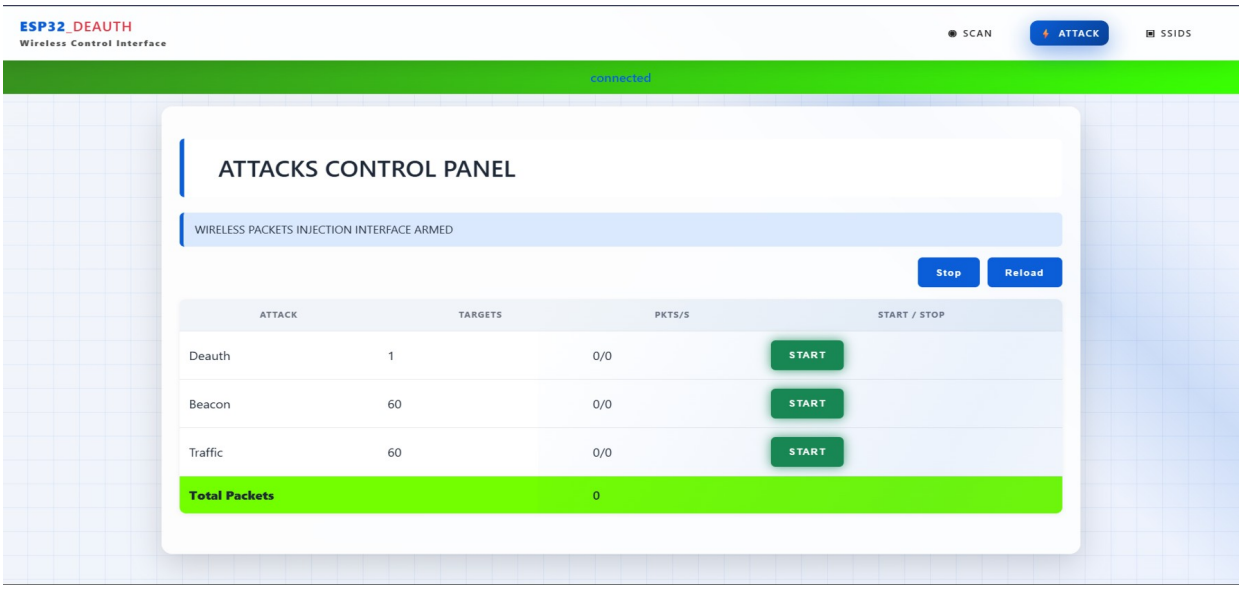


Fig 7.3: Attack Execution Phase

The above figure shows the Attack Control Panel, which is responsible for executing wireless penetration testing operations. The panel includes options for different attack types such as deauthentication, beacon flooding, and traffic injection.The system displays parameters such as number of targets and packet transmission rate (packets per second). Users can start or stop attacks using dedicated control buttons. During testing, the ESP32 successfully initiated packet injection operations, and the system displayed real-time status updates.

The interface also includes a total packet counter, which reflects the number of transmitted packets during the attack. The control panel ensures that attack execution is structured and user-controlled, preventing unintended operations.The system maintains controlled execution by regulating packet transmission rates, ensuring stability of the ESP32 module during continuous attack operations. Additionally, safeguards are implemented to prevent overlapping attack commands, thereby maintaining system reliability and preventing unintended interference with ongoing processes. The real-time feedback mechanism enhances user awareness by continuously updating attack status and execution metrics.

The attack panel also provides flexibility in selecting multiple targets simultaneously, enabling batch execution of attack operations for broader testing scenarios. The system dynamically adapts to channel conditions and ensures that packet injection is aligned with the selected network parameters. This enhances the effectiveness of the attack while maintaining synchronization with real-time network behavior. This result confirms that the Attack Simulation Module is functioning correctly and is capable of executing controlled wireless attacks.

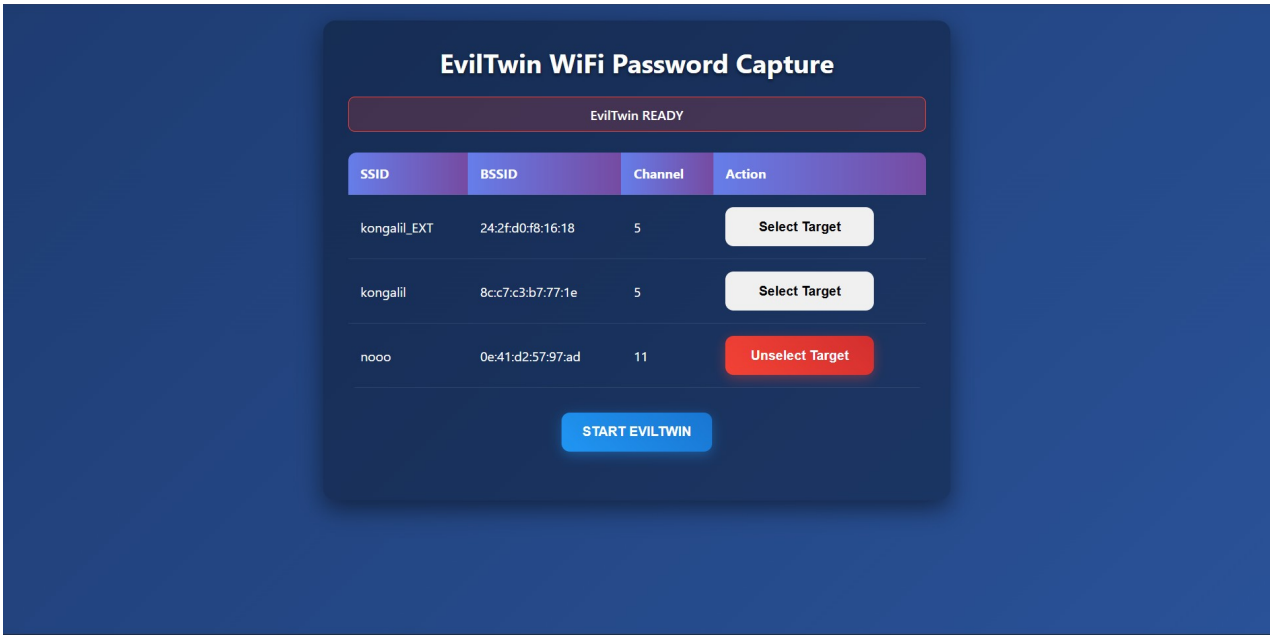


Fig 7.4: Evil Twin Attack Setup

The above figure shows the configuration interface for the Evil Twin attack. The system allows users to select a target access point based on SSID, BSSID, and channel information.Once a target is selected, the system prepares to launch a rogue access point that mimics the original network. The interface indicates readiness status and provides an option to start the Evil Twin attack.During testing, the system successfully identified multiple networks and allowed target selection. This setup phase is critical for performing credential capture attacks. This result demonstrates that the system can configure and prepare advanced wireless attacks using real-time network data

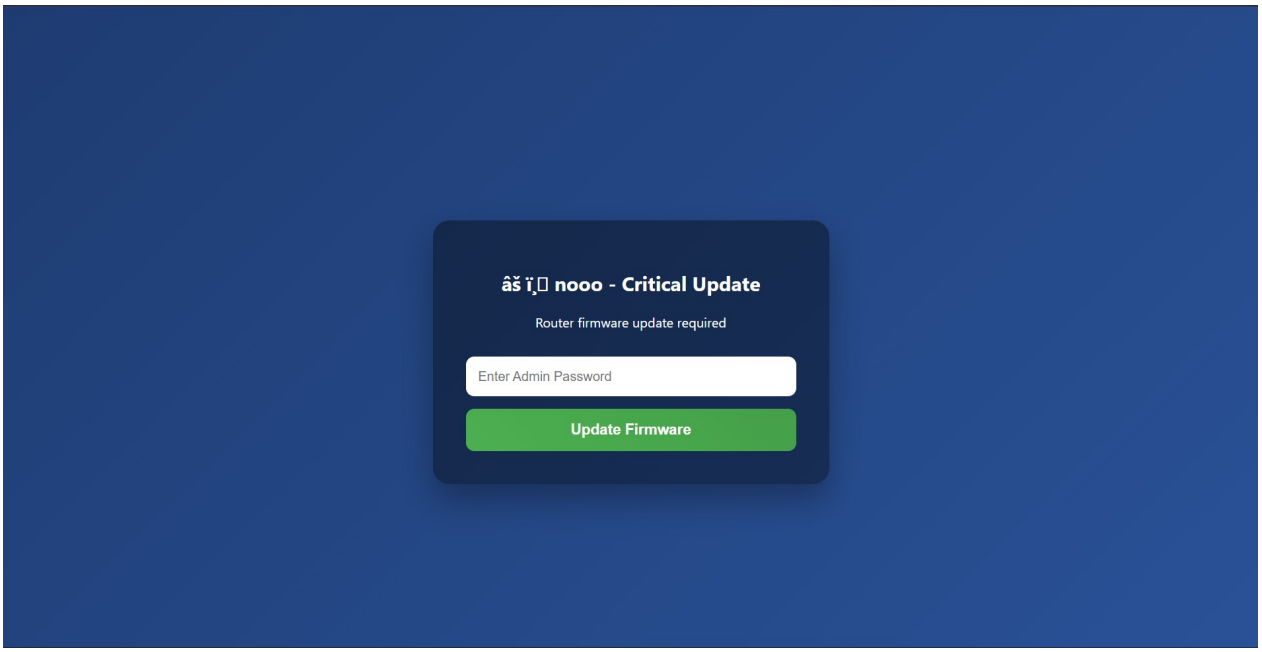


Fig 7.5: Credential Capture Interface

The above figure shows the phishing interface used in the Evil Twin attack. The system presents a fake login page that prompts users to enter network credentials under the pretext of a firmware update or authentication requirement.The interface includes an input field for password entry and a submission button. This page is designed to mimic legitimate network behavior and trick users into providing credentials.

During testing, the system successfully displayed the phishing page and captured input data entered by the user. The captured credentials were transmitted to the backend system for storage and analysis.This result confirms that the system can perform credential harvesting in a controlled penetration testing environment.

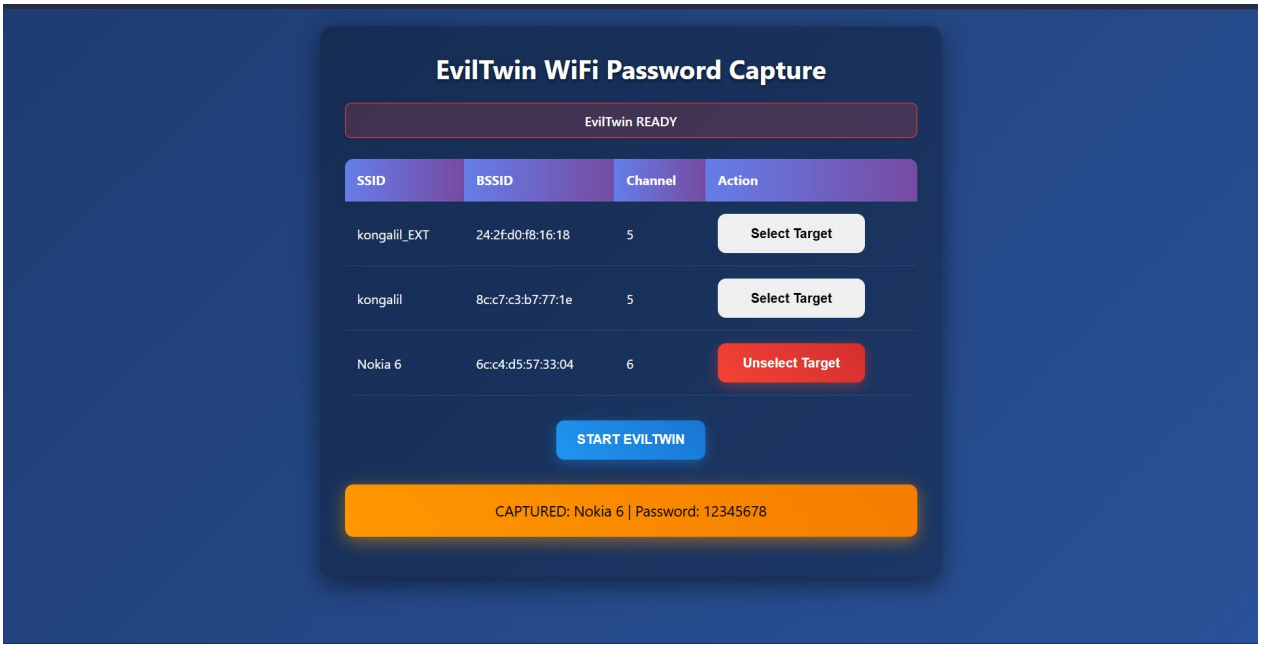


Fig 7.6: Successful Credential Capture Output

The above figure shows the successful capture of user credentials during the Evil Twin attack. The system displays the captured SSID along with the corresponding password entered by the user.

The captured data is highlighted and stored within the system, confirming successful execution of the attack. This demonstrates the vulnerability of users to social engineering attacks when connected to rogue access points.The captured credentials are securely logged within the backend system along with associated metadata such as timestamp, target SSID, and device identification parameters. This enables further forensic analysis and validation of attack success. The result highlights the effectiveness of social engineering-based attacks in wireless environments and emphasizes the need for user awareness and secure authentication mechanisms to prevent credential compromise. The system successfully integrates scanning, attack execution, and data capture into a unified workflow, providing a complete demonstration of wireless attack scenarios.

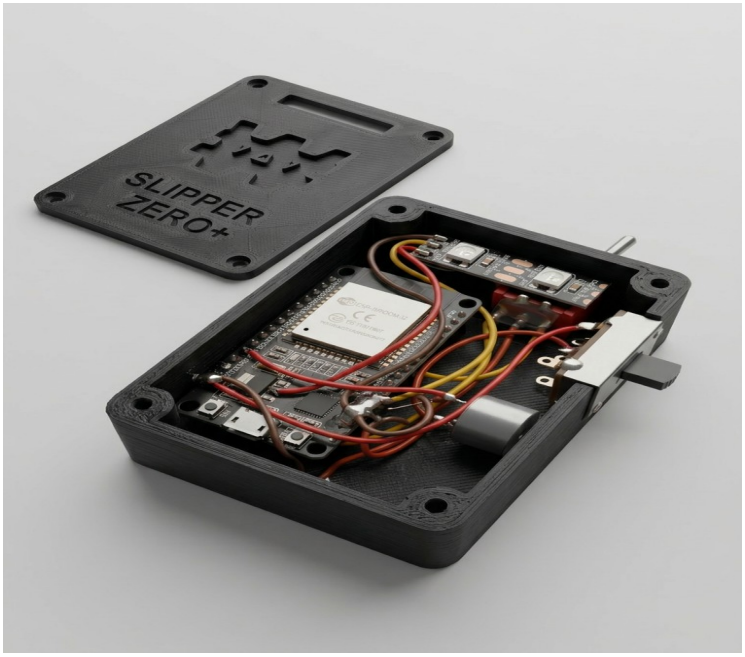


Fig 7.7: Hardware Implementation of Slipper Zero+ Device

The above figure illustrates the physical hardware implementation of the proposed Slipper Zero Plus system. The device is developed using a compact 3D-printed enclosure that houses all electronic components in an integrated and portable form factor. The enclosure is designed with a removable top cover, allowing visibility and accessibility to internal components for testing and maintenance purposes.The core of the hardware setup is the ESP32 NodeMCU development board, which acts as the central processing unit responsible for executing wireless scanning, packet monitoring, intrusion detection, and controlled attack operations. The ESP32 is positioned centrally within the enclosure and is connected to various peripheral components through manually soldered wiring connections.The system includes an integrated LED module consisting of multiple LEDs that provide real-time visual feedback regarding system states such as scanning activity, attack execution, and alert generation. Additionally, a cylindrical buzzer is incorporated to provide audible alerts when suspicious activity or intrusion patterns are detected by the system. These indicators enhance the usability of the device by allowing users to understand system behavior without relying solely on the web dashboard.A metal toggle switch is mounted on the side of the enclosure to control power or operational modes of the system. The device also includes push buttons for boot and reset operations, ensuring proper control during initialization and firmware execution. The power supply is provided through a micro-USB interface connected to the ESP32 board, ensuring stable and consistent operation.The successful integration of all hardware components within a compact enclosure confirms the feasibility of developing a portable wireless security analysis device.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This project was undertaken to address the growing need for an accessible, portable, and real-time wireless security testing platform capable of performing both penetration testing and intrusion monitoring without relying on complex external tools. With the rapid expansion of Wi-Fi-based communication systems, networks remain vulnerable to attacks such as deauthentication, rogue access points, packet sniffing, and credential harvesting. Traditional tools often require specialized operating systems, external adapters, and command-line expertise, limiting their usability in educational and field environments.The proposed Slipper Zero Plus system was successfully designed and implemented as a hybrid embedded and web-based platform combining ESP32 hardware with a Python Flask backend and a browser-based dashboard interface. The ESP32 module performs core wireless operations such as network scanning, packet sniffing in promiscuous mode, and attack execution including deauthentication and Evil Twin simulations. These operations are controlled and monitored through a backend server developed using Flask, which acts as the central coordination layer of the system.

The frontend of the system is implemented using HTML, CSS, and JavaScript, providing an interactive and user-friendly web dashboard. This interface allows users to visualize nearby wireless networks, monitor packet activity, select targets, and execute attacks through graphical controls rather than command-line tools. Real-time communication between the ESP32 device, backend server, and dashboard is achieved using WebSocket and HTTP-based communication mechanisms, ensuring continuous updates of network status, logs, and attack results.During system operation, the ESP32 captures wireless frames and forwards relevant data to the Flask backend, where it is processed, stored, and transmitted to the frontend dashboard. The dashboard dynamically updates network information such as SSIDs, signal strength, encryption types, connected stations, and detected anomalies. Users can initiate operations such as scanning, deauthentication attacks, and Evil Twin deployment directly from the interface, with results displayed in real time.

The system successfully demonstrated offensive capabilities including targeted deauthentication attacks and rogue access point deployment for credential capture simulation. At the same time, it provided defensive insights by identifying weak encryption mechanisms, duplicate SSIDs, and abnormal packet behavior. The Evil Twin module further demonstrated how users can be deceived into connecting to malicious networks, highlighting real-world wireless security risks.

The integration of embedded firmware, backend processing, and web-based visualization creates a complete full-stack cybersecurity solution that is both powerful and easy to use. The system eliminates the dependency on traditional penetration testing environments while maintaining core functionalities required for wireless auditing and security analysis.

The results obtained from implementation and testing confirm that the system achieves its intended objectives of providing a standalone, low-cost, and real-time Wi-Fi security toolkit. The modular design ensures scalability and allows future enhancements such as advanced analytics, automated detection, and support for additional wireless protocols.

Overall, the Slipper Zero Plus system represents a practical and efficient approach to wireless penetration testing and security education, combining hardware-level capabilities with modern web technologies to deliver a comprehensive and user-friendly cybersecurity platform.

8.2 Future Scope

The Slipper Zero Plus system establishes a strong foundation for wireless penetration testing and intrusion detection using an embedded ESP32 platform integrated with a Python Flask backend and a web-based dashboard. While the current implementation successfully demonstrates core functionalities, there are several opportunities for enhancement to improve system intelligence, scalability, automation, and real-world applicability. The following future directions provide a comprehensive roadmap for advancing the system into a more powerful and versatile wireless security framework.

1. AI and Machine Learning–Based Threat Detection

The current system utilizes rule-based detection mechanisms and predefined thresholds to identify suspicious wireless activities such as abnormal packet rates or rogue access points. Although effective for known attack patterns, this approach has limitations when dealing with unknown or evolving threats.Future enhancements can integrate Artificial Intelligence (AI) and Machine Learning (ML) techniques within the Python Flask backend to enable intelligent and adaptive threat detection. Machine learning models can be trained using datasets containing both normal and malicious wireless traffic patterns. These models can analyze features such as packet frequency, timing intervals, frame types, signal variations, and communication behavior to identify anomalies.Advanced techniques such as supervised learning (e.g., Random Forest, Support Vector Machines), unsupervised learning (e.g., clustering and anomaly detection), and deep learning (e.g., neuralnetworks and LSTM models for sequential data) can be implemented. These models can detectzero-day attacks, polymorphic behavior, and unknown intrusion techniques that are not covered by static rules.By continuously learning from new data, the system can dynamically update its detection logic, improving accuracy and reducing false positives.

This transformation will enable the system to evolve into an intelligent wireless intrusion detection system capable of adapting to modern cybersecurity threats.

2. Multi-Protocol Wireless Support

The current implementation is focused primarily on Wi-Fi (IEEE 802.11) networks. However, modern environments include a wide range of wireless communication technologies used in IoT devices, smart systems, and industrial applications.Future versions of the system can be extended to support additional wireless protocols such as Bluetooth Low Energy (BLE), ZigBee, NFC, RFID, and LoRa. Each of these protocols has its own communication characteristics, vulnerabilities, and attack surfaces.By integrating support for multiple protocols, the system can provide a more comprehensive security analysis platform.For example, Bluetooth support can enable detection of insecure pairing mechanisms, ZigBee integration can allow monitoring of smart home networks, and RFID/NFC analysis can be used for access control security testing. This expansion will require modifications in both hardware capabilities and firmware design, potentially incorporating additional radio modules.By supporting multiple wireless technologies, the system can evolve from a Wi-Fi testing tool into a universal wireless cybersecurity framework suitable for IoT environments, smart cities, and industrial networks.

3. Advanced Dashboard and Data Visualization

The existing web dashboard provides real-time monitoring and control using HTML, CSS, and JavaScript. While it effectively displays network data and alerts, future improvements can enhance the visualization and analytical capabilities of the interface. Advanced frontend features can include interactive charts, dynamic graphs, and visual analytics tools using libraries such as Chart.js, D3.js, or WebGL-based visualization frameworks. These features can display packet trends, signal strength variations, network density, and attack patterns over time.The system can also incorporate historical data storage and timeline-based analysis, allowing users to review past events and identify long-term patterns in network behavior. Heat maps can be used to visualize signal distribution, while attack replay functionality can demonstrate how specific attacks were executed and detected. Additionally, integrating real-time filtering and customizable dashboards can allow users to focus on specific parameters or threat types, improving analytical efficiency.Furthermore, automated report generation features can be implemented to export scan results, alerts, and logs in formats such as PDF, CSV, or JSON. These reports can be used for documentation, auditing, and academic evaluation purposes. The inclusion of role-based access control and secure authentication mechanisms can further enhance the reliability and usability of the dashboard in multi-user environments. Such enhancements will significantly improve user experience and make the system more suitable for professional analysis and cybersecurity research.

4. Automated Intrusion Response System

Currently, the system primarily focuses on detection and user-controlled attack simulation. In future implementations, the system can be enhanced to include automated response mechanisms that actively mitigate detected threats.When suspicious activity is identified, the system can automatically trigger predefined actions such as blocking specific MAC addresses, disconnecting malicious clients, isolating rogue access points, or switching network channels to avoid interference. These actions can be controlled through the Flask backend, enabling real-time decision-making.The system can also implement automated alert notifications through email, mobile applications, or messaging platforms, ensuring that users are immediately informed of potential threats. Integration with firewall rules or network access control mechanisms can further strengthen system security.By incorporating automated response capabilities, the system can evolve from a passive monitoring tool into an active Intrusion Prevention System (IPS), capable of both detecting and responding to attacks without requiring manual intervention.

5. Cloud Integration and Remote Management

The current system operates as a standalone platform with local processing and dashboard access. Future enhancements can include cloud integration to enable centralized data storage, remote monitoring, and distributed system management.The Python Flask backend can be extended to communicate with cloud services for storing logs, sharing threat intelligence, and synchronizing data across multiple devices. This will allow multiple Slipper Zero Plus units deployed in different locations to operate collaboratively.Remote access features can be implemented to allow administrators to monitor and control the system from any location using secure web interfaces. Cloud-based dashboards can provide a global view of network activity, enabling large-scale monitoring.Additionally, cloud integration can support automatic updates, model training for AI-based detection, and data backup, ensuring system reliability and scalability. This enhancement will make the platform suitable for enterprise-level deployments and large network environments.

6. Enhanced Security and Authentication Mechanisms

As the system evolves, it becomes important to secure access to the dashboard and communication channels. Future versions can implement advanced authentication and security mechanisms to protect sensitive data and system operations.Features such as user authentication, role-based access control, and secure login systems can be added to restrict access to authorized users.Communication between the ESP32, Flask backend, and dashboard can be secured using encryption protocols such as HTTPS and secure WebSockets.Additional measures such as token-based authentication, session management, and logging of user activities can further enhance system security.These improvements will ensure that the system can be safely deployed in professional and enterprise environments without compromising data integrity.

7. Hardware Enhancement and Performance Optimization

While the ESP32 provides a cost-effective and portable solution, it has limitations in terms of processing power and memory. Future implementations can explore more advanced hardware platforms such as Raspberry Pi or dedicated Wi-Fi adapters to support additional functionalities. These platforms offer higher computational capabilities, enabling the execution of more complex algorithms and large-scale data processing tasks.Enhanced hardware can enable features such as multi-channel scanning, deeper packet inspection, improved attack execution, and support for larger datasets. External antennas can also be integrated to increase signal range and improve detection accuracy, especially in dense or large network environments. Additionally, the use of multiple wireless interfaces can allow simultaneous monitoring of different channels, improving coverage and detection efficiency.Optimization techniques can be applied to improve system performance, reduce latency, and enhance real-time processing capabilities. This includes efficient memory management, task scheduling, and the use of lightweight data structures to handle high volumes of network traffic. Hardware acceleration techniques and optimized firmware can further improve system responsiveness.Furthermore, integrating dedicated co-processors or edge computing modules can offload intensive tasks such as packet analysis and intrusion detection, thereby reducing the burden on the main controller. Power management strategies can also be implemented to ensure energy-efficient operation, especially for portable or battery-powered deployments.These improvements will make the system more robust, scalable, and suitable for large-scale deployments, enterprise environments, and advanced cybersecurity research applications.

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